

Electromagnetic Scattering

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Chapter Overview

This chapter provides an overview of electromagnetic scattering, a fundamental phenomenon that occurs when a wave encounters an obstacle. We will begin by defining the concept of scattering cross-section, a key parameter used to quantify the strength and angular distribution of scattered fields. The chapter will then delve into the analytical solutions for scattering by canonical objects, starting with two-dimensional problems involving PEC cylinders, and then extending the analysis to three-dimensional scattering by dielectric spheres, commonly known as Mie theory.

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Cross Sections

In the analysis of scattering problems, the concept of cross section is frequently employed to quantitatively characterize how an object interacts with and scatters electromagnetic waves in the far field. In three dimensions, a cross section measures an effective area that intercepts and reradiates incident energy; in a two-dimensional problem invariant along one axis, the corresponding quantity is an effective transverse width, with power understood per unit invariant length. In this chapter, various types of cross sections are defined. These quantities prove particularly useful when comparing the scattering behavior of different objects or materials.

Unless stated otherwise, the exterior medium in this chapter is homogeneous, isotropic, and lossless, with real positive ϵ and μ , wavenumber $k = \omega\sqrt{\mu\epsilon} > 0$, and wave impedance $\eta = \sqrt{\mu/\epsilon} > 0$. This assumption is needed below when the real, time-averaged radial power density is written as $|E|^2/(2\eta)$. A lossy host requires $\text{Re}(\vec{E} \times \vec{H}^*)/2$ and a correspondingly modified normalization.

A scattering problem typically involves determining the field scattered by an obstacle. The incident field is often represented as a plane wave propagating in the $\hat{i}$ direction:

$$\vec{E}_i = \vec{E}_0 e^{-ik\hat{i}\cdot\vec{r}}, \quad \hat{i} \cdot \vec{E}_0 = 0. \quad (4.1.1)$$

Cross Sections (cont.)

Physically, this plane-wave representation approximates the far field produced by a radar antenna located at a large distance from the target. In this configuration, the wavefronts arriving at the target are effectively planar, and the observation (receiving) antenna is assumed to be in the target's far-field zone. A localized three-dimensional target then produces a radiating spherical wave; the ideal two-dimensional, translation-invariant model used for an infinite cylinder instead produces a cylindrical wave. The scattered field is defined in either case by

$$\vec{E}_s = \vec{E}_t - \vec{E}_i. \quad (4.1.2)$$

The interface matching conditions are imposed on the total field, and the scattered field must additionally satisfy a no-incoming-wave boundary condition. This radiation condition is prescribed, not derived from matching alone. Section 3.1.3 proves uniqueness for the stated homogeneous lossless exterior class.

The three-dimensional e^{-ikr}/r factor follows from the finite-source Green-function expansion in Chapter 3. The two-dimensional factor $e^{-ik\rho}/\sqrt{\rho}$ follows from the outgoing cylindrical solution and its asymptotics derived in Section B.5.1. The scattering-amplitude formula below denotes the leading far-field term, not an exact field

Cross Sections (cont.)

at finite distance. The 4π and 2π radar-cross-section factors are normalization conventions motivated by isotropic reradiation in three and two dimensions, respectively. In the far field, let $E_0 = |\vec{E}_0|$, and let $\vec{f}(\hat{s}, \hat{i})$ denote the vector scattering-amplitude function in observation direction $\hat{s}$ for incidence direction $\hat{i}$. With these quantities defined, the scattered field takes the form

$$\vec{E}_s = \begin{cases} E_0 \frac{e^{-ikr}}{r} \vec{f}(\hat{s}, \hat{i}) & (3D), \\ E_0 \frac{e^{-ik\rho}}{\sqrt{\rho}} \vec{f}(\hat{s}, \hat{i}) & (2D). \end{cases} \quad (4.1.3)$$

From (1.5.7), the incident and scattered power flux densities are evaluated by noting that each field is locally a plane wave, so its magnetic field follows from Faraday's law $\nabla \times \vec{E} = -i\omega\mu\vec{H}$. For the incident wave (4.1.1), $\nabla \times \vec{E}_i = -ik\hat{i} \times \vec{E}_i$, and since $k/(\omega\mu) = 1/\eta$,

$$\vec{H}_i = \frac{1}{\eta} \hat{i} \times \vec{E}_i.$$

Cross Sections (cont.)

Substituting this into the Poynting vector and applying the identity $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$,

$$\begin{aligned}\vec{S}_i &= \frac{1}{2} \vec{E}_i \times \vec{H}_i^* = \frac{1}{2\eta} \vec{E}_i \times (\hat{i} \times \vec{E}_i^*) \\ &= \frac{1}{2\eta} \left[\hat{i} (\vec{E}_i \cdot \vec{E}_i^*) - \vec{E}_i^* (\vec{E}_i \cdot \hat{i}) \right].\end{aligned}$$

The plane wave is transverse, $\hat{i} \cdot \vec{E}_i = 0$, so the second term vanishes; writing $|\vec{E}_i|^2 \equiv \vec{E}_i \cdot \vec{E}_i^*$, only the longitudinal part remains. The scattered field is locally plane in the far zone, whether its radial spreading is spherical in 3D or cylindrical in 2D, so the same steps apply with $\hat{i} \rightarrow \hat{s}$ and $\vec{E}_i \rightarrow \vec{E}_s$. This gives

$$\vec{S}_i = \frac{1}{2} (\vec{E}_i \times \vec{H}_i^*) = \frac{|\vec{E}_i|^2}{2\eta} \hat{i}, \quad \vec{S}_s = \frac{1}{2} (\vec{E}_s \times \vec{H}_s^*) = \frac{|\vec{E}_s|^2}{2\eta} \hat{s}. \quad (4.1.4)$$

The differential scattering cross section is defined as:

$$\sigma_d(\hat{s}, \hat{i}) = \begin{cases} \lim_{r \rightarrow \infty} r^2 \frac{|\vec{S}_s|}{|\vec{S}_i|} & (3D) \\ \lim_{\rho \rightarrow \infty} \rho \frac{|\vec{S}_s|}{|\vec{S}_i|} & (2D) \end{cases} = |\vec{f}(\hat{s}, \hat{i})|^2. \quad (4.1.5)$$

Cross Sections (cont.)

To evaluate this ratio, note that the incident plane wave (4.1.1) has $|\vec{E}_i| = E_0$, so that (1.5.7) gives

$$|\vec{S}_i| = \frac{|\vec{E}_i|^2}{2\eta} = \frac{E_0^2}{2\eta}.$$

For the scattered field, $\hat{s}$ is a unit vector and the lossless radial phase factor has unit modulus ($|e^{-ikr}| = |e^{-ik\rho}| = 1$), so the far-field form yields

$$|\vec{E}_s|^2 = \frac{E_0^2}{r^2} |\vec{f}|^2 \quad (3D), \quad |\vec{E}_s|^2 = \frac{E_0^2}{\rho} |\vec{f}|^2 \quad (2D),$$

where $|\vec{f}|^2 = \vec{f} \cdot \vec{f}^*$. Hence

$$|\vec{S}_s| = \frac{|\vec{E}_s|^2}{2\eta} = \begin{cases} \frac{E_0^2 |\vec{f}|^2}{2\eta r^2} & (3D), \\ \frac{E_0^2 |\vec{f}|^2}{2\eta \rho} & (2D). \end{cases}$$

Cross Sections (cont.)

Forming the ratio, the $1/(2\eta)$ factors and the radial decay cancel,

$$r^2 \frac{|\vec{S}_s|}{|\vec{S}_i|} = |\vec{f}|^2 \quad (3D), \quad \rho \frac{|\vec{S}_s|}{|\vec{S}_i|} = |\vec{f}|^2 \quad (2D),$$

which is already independent of r (or ρ), so the limit is immediate, establishing the last equality, $\sigma_d = |\vec{f}|^2$, in (4.1.5). The dimensions follow directly from the far-field normalization in (4.1.3): $\vec{f}$ has dimensions of length in 3D and of the square root of length in 2D. Consequently, σ_d has dimensions of area in 3D but of length in 2D. The bistatic radar cross section (RCS) is then defined as:

$$\sigma_{bi}(\hat{s}, \hat{i}) = \begin{cases} 4\pi\sigma_d(\hat{s}, \hat{i}) & (3D), \\ 2\pi\sigma_d(\hat{s}, \hat{i}) & (2D). \end{cases} \quad (4.1.6)$$

In two dimensions, σ_{bi} is also referred to as the echo width.

In 3D, the bistatic RCS represents a hypothetical area that, if illuminated by the incident power density and reradiating that power isotropically, would produce the same received power density as the actual target. In 2D, the analogous echo width is a hypothetical transverse width, not an area; incident and scattered powers are understood per unit

Cross Sections (cont.)

invariant cylinder length. Both quantities are defined to be independent of the distance between the radar and the target, but they strongly depend on the incidence angle, observation angle, polarization, frequency, and the target's material and shape. Specifically, the monostatic or backscattering RCS refers to the scenario where the radar transmitter and receiver are co-located, thereby measuring the power reflected directly back toward the source:

$$\sigma_{mono}(\hat{i}) = \sigma_{bi}(-\hat{i}, \hat{i}). \quad (4.1.7)$$

Finally, the total scattering cross section, which quantifies the amount of incident power scattered by an object in all directions, is defined as:

$$\sigma_{sca} = \begin{cases} \int_{4\pi} \sigma_d d\Omega = \int_{4\pi} r^2 \frac{|\vec{S}_s|}{|\vec{S}_i|} d\Omega & (3D), \\ \int_{2\pi} \sigma_d d\phi = \int_{2\pi} \rho \frac{|\vec{S}_s|}{|\vec{S}_i|} d\phi & (2D), \end{cases} \quad (4.1.8)$$

where $d\Omega$ represents the differential solid angle.

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Cylindrical Waves

This section focuses on the wave and Helmholtz equations in cylindrical coordinates and on the corresponding cylinder-scattering problem; spherical waves and sphere scattering are treated separately in Section 4.4. As noted in Section 1.4, in a source-free region, the scalar Helmholtz equation (1.4.16) applies to all three Cartesian components. Of the physical components resolved in the cylindrical basis, only the z component has a position-independent unit vector and therefore satisfies that scalar equation directly, without curvature-induced coupling to another cylindrical component. The Cartesian x and y components still satisfy scalar Helmholtz equations; they are simply not the physical radial and azimuthal components. In spherical coordinates, none of the local unit vectors are constant, necessitating the additional analysis in Section 4.4.

Cylindrical Wave Solution

For the z component that scalar equation is the cylindrical Helmholtz equation

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \psi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{\partial^2 \psi}{\partial z^2} + k^2 \psi = 0. \quad (4.2.1)$$

This equation can be solved using the method of separation of variables, by assuming a solution of the form:

$$\psi = B(\rho)\Phi(\phi)Z(z). \quad (4.2.2)$$

Substituting this into (4.2.1) yields:

$$\frac{1}{B\rho} \frac{d}{d\rho} \left(\rho \frac{dB}{d\rho} \right) + \frac{1}{\Phi\rho^2} \frac{d^2\Phi}{d\phi^2} + \frac{1}{Z} \frac{d^2Z}{dz^2} + k^2 = 0. \quad (4.2.3)$$

Since the variable z appears only in the third term of (4.2.3), we can separate it, leading to an equation similar to the rectangular case:

$$\frac{d^2Z}{dz^2} + k_z^2 Z = 0. \quad (4.2.4)$$

Cylindrical Wave Solution (cont.)

For $k_z \neq 0$, the elementary solutions for Z are:

$$Z = e^{\pm ik_z z}. \quad (4.2.5)$$

At $k_z = 0$ these two exponentials coalesce; the full solution of $Z'' = 0$ is $Z = A + Bz$. The two-dimensional cylinder-scattering problem below is invariant along z and selects the constant solution. Now, let

$$k_\rho^2 = k^2 - k_z^2. \quad (4.2.6)$$

Multiplying (4.2.3) by ρ^2 , we obtain:

$$\frac{\rho}{B} \frac{d}{d\rho} \left(\rho \frac{dB}{d\rho} \right) + \frac{1}{\Phi} \frac{d^2\Phi}{d\phi^2} + k_\rho^2 \rho^2 = 0. \quad (4.2.7)$$

Similarly, the middle term of (4.2.7), $\frac{1}{\Phi} \frac{d^2\Phi}{d\phi^2}$, depends only on ϕ , whereas the remaining terms depend only on ρ . Since their sum must vanish for all ρ and ϕ independently, each part must equal a constant. We write the azimuthal part as a negative constant,

$$\frac{1}{\Phi} \frac{d^2\Phi}{d\phi^2} = -\nu^2,$$

Cylindrical Wave Solution (cont.)

so that the radial remainder equals $+\nu^2$, that is,

$$\frac{\rho}{B} \frac{d}{d\rho} \left(\rho \frac{dB}{d\rho} \right) + k_\rho^2 \rho^2 = \nu^2.$$

The negative sign is required for Φ to be single-valued over 2π : it gives the oscillatory solution $\Phi = e^{\pm i\nu\phi}$, whereas a positive constant would yield real exponentials that cannot be periodic. The azimuthal part thus reproduces the equation for Φ in (4.2.8),

$$\frac{d^2\Phi}{d\phi^2} + \nu^2\Phi = 0, \quad (4.2.8)$$

which has the elementary solution of:

$$\Phi = e^{\pm i\nu\phi}. \quad (4.2.9)$$

By the same periodicity, $\nu = n$ is an integer constant. Consequently, (4.2.7) simplifies to:

$$\frac{d^2B}{d\rho^2} + \frac{1}{\rho} \frac{dB}{d\rho} + \left(k_\rho^2 - \frac{\nu^2}{\rho^2} \right) B = 0. \quad (4.2.10)$$

Cylindrical Wave Solution (cont.)

This is known as the Bessel equation of order ν . For $k_\rho \neq 0$, its solutions are the Bessel functions described below. The cutoff case $k_\rho = 0$ must be taken separately: the radial equation is then an Euler equation with

$$\begin{aligned} B(\rho) &= C\rho^\nu + D\rho^{-\nu}, & \nu > 0, \\ B(\rho) &= C + D \ln \rho, & \nu = 0. \end{aligned}$$

Thus one must not obtain a cutoff basis by directly substituting $k_\rho = 0$ into $J_\nu(k_\rho\rho)$ and $Y_\nu(k_\rho\rho)$. The propagating wave and cylinder-scattering formulas below use real $k_\rho > 0$. For $\nu \geq 0$, the solution to (4.2.10) that remains finite at $\rho = 0$ is known as the Bessel function of the first kind:

$$J_\nu(k_\rho\rho) = \sum_{m=0}^{\infty} \frac{(-1)^m (k_\rho\rho/2)^{\nu+2m}}{m! \Gamma(m + \nu + 1)}. \quad (4.2.11)$$

When ν is a nonnegative integer, $\Gamma(m + \nu + 1) = (m + \nu)!$, recovering the factorial form. Because the differential equation depends on ν^2 , we may take $\nu \geq 0$. For

Cylindrical Wave Solution (cont.)

noninteger $\nu > 0$, a second independent solution, $J_{-\nu}(k_\rho \rho)$, exists but diverges at $\rho = 0$. If ν is an integer, denoted by n , then:

$$J_{-n}(k_\rho \rho) = (-1)^n J_n(k_\rho \rho). \quad (4.2.12)$$

A second independent solution, called the Neumann function, is constructed as:

$$Y_n(k_\rho \rho) = \lim_{\nu \rightarrow n} \frac{J_\nu(k_\rho \rho) \cos(\nu\pi) - J_{-\nu}(k_\rho \rho)}{\sin(\nu\pi)}. \quad (4.2.13)$$

The Hankel functions of the first and second kind are combinations of the Bessel and Neumann functions:

$$H_n^{(1)}(k_\rho \rho) = J_n(k_\rho \rho) + iY_n(k_\rho \rho), \quad (4.2.14)$$

$$H_n^{(2)}(k_\rho \rho) = J_n(k_\rho \rho) - iY_n(k_\rho \rho). \quad (4.2.15)$$

For each fixed nonnegative integer n and real $k_\rho \rho \rightarrow +\infty$, Section B.5.1 derives the following leading Hankel forms; their error constants depend on n :

$$H_n^{(1)}(k_\rho \rho) \approx \sqrt{\frac{2}{\pi k_\rho \rho}} e^{i(k_\rho \rho - n\pi/2 - \pi/4)}, \quad (4.2.16)$$

Cylindrical Wave Solution (cont.)

$$H_n^{(2)}(k_\rho \rho) \approx \sqrt{\frac{2}{\pi k_\rho \rho}} e^{-i(k_\rho \rho - n\pi/2 - \pi/4)}. \quad (4.2.17)$$

For real $k_\rho > 0$, these asymptotics, with the $e^{i\omega t}$ time convention, represent incoming and outgoing traveling cylindrical waves, respectively. If k_ρ is imaginary or genuinely complex, the radial behavior is evanescent or attenuated and this traveling-wave phase interpretation must be replaced by the corresponding analytic continuation. A few remarks justify the construction above. The relation $J_{-n} = (-1)^n J_n$ shows that, for integer order, J_n and J_{-n} are not independent: their Wronskian

$$W_x\{J_\nu, J_{-\nu}\}(x) = -\frac{2 \sin(\nu\pi)}{\pi x} \quad \text{with } x = k_\rho \rho,$$

where $W_x\{f, g\} = f(dg/dx) - (df/dx)g$, vanishes as $\nu \rightarrow n$, so a single power series cannot span the two-dimensional solution space of the second-order equation (4.2.10). The Neumann function (4.2.13) restores the missing second solution. At integer order its defining quotient is indeterminate, since

$$J_n \cos(n\pi) - J_{-n} = (-1)^n J_n - (-1)^n J_n = 0, \quad \sin(n\pi) = 0,$$

Cylindrical Wave Solution (cont.)

a $0/0$ form; the limit is evaluated by L'Hôpital's rule, differentiating numerator and denominator with respect to ν . Section B.5.1 derives the integer-order positive-real-axis forms (4.2.16) and (4.2.17), including their phase and normalization. General-order complex-sector versions remain explicitly identified reference results [9, 1]. The traveling-wave assignment is checked directly in Appendix B, in the discussion following (B.3.7): restoring the suppressed factor $e^{i\omega t}$ gives the space-time phases $e^{i(\omega t \pm k_\rho \rho)}$, whose constant-phase surfaces move inward for $H_n^{(1)}$ and outward for $H_n^{(2)}$. Hence $H_n^{(2)}$ is the appropriate choice for a scattered field radiating away from the cylinder.

Cylindrical Wave Transformation

Consider a z-polarized plane wave propagating in the x-direction:

$$\vec{E} = \hat{z} E_0 e^{-ikx} = \hat{z} E_0 e^{-ik\rho \cos \phi}. \quad (4.2.18)$$

We can expand this exponential term into an infinite sum of cylindrical waves:

$$e^{-ik\rho \cos \phi} = \sum_{n=-\infty}^{\infty} p_n^{(\text{cyl})} J_n(k\rho) e^{in\phi}. \quad (4.2.19)$$

To determine the temporary plane-wave coefficients $p_n^{(\text{cyl})}$, first derive the nontrivial integral rather than assume it. The Bessel generating function is

$$\exp\left[\frac{z}{2}(t - t^{-1})\right] = \sum_{n=-\infty}^{\infty} J_n(z) t^n.$$

Setting $t = -ie^{i\phi}$ gives $t - t^{-1} = -2i \cos \phi$ and therefore

$$e^{-iz \cos \phi} = \sum_{n=-\infty}^{\infty} i^{-n} J_n(z) e^{in\phi}.$$

Cylindrical Wave Transformation (cont.)

Multiplying this generating-function identity by $e^{-im\phi}$ and using Fourier orthogonality gives the first of the two identities used below; the second is the orthogonality relation itself:

$$\int_0^{2\pi} e^{-i(k\rho \cos \phi + m\phi)} d\phi = 2\pi i^{-m} J_m(k\rho), \quad (4.2.20)$$

$$\int_0^{2\pi} e^{i(m-n)\phi} d\phi = 2\pi \delta_{mn}. \quad (4.2.21)$$

Multiplying (4.2.19) by $e^{-im\phi}$ and integrating over ϕ from 0 to 2π projects out a single term: the left-hand side is evaluated by the first identity, while the second identity collapses the right-hand sum to its $n = m$ term,

$$\begin{aligned} \int_0^{2\pi} e^{-ik\rho \cos \phi} e^{-im\phi} d\phi &= \sum_{n=-\infty}^{\infty} p_n^{(\text{cyl})} J_n(k\rho) \\ &\quad \times \int_0^{2\pi} e^{i(n-m)\phi} d\phi, \end{aligned}$$

Cylindrical Wave Transformation (cont.)

that is, $2\pi i^{-m} J_m(k\rho) = 2\pi p_m^{(\text{cyl})} J_m(k\rho)$ for every ρ . The function $J_m(k\rho)$ is not identically zero, so choose any ρ outside its isolated zero set. Cancellation there gives a coefficient identity that is independent of ρ and therefore applies to the entire expansion:

$$p_m^{(\text{cyl})} = i^{-m}. \quad (4.2.22)$$

Thus, we obtain the following expansion:

$$E_z = E_0 e^{-ikx} = E_0 e^{-ik\rho \cos \phi} = E_0 \sum_{n=-\infty}^{\infty} i^{-n} J_n(k\rho) e^{in\phi}. \quad (4.2.23)$$

This process is known as the cylindrical wave transformation, which expands a plane wave into a sum of cylindrical waves.

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Scattering from PEC Cylinders

This section examines the scattering of a plane wave by a PEC cylinder of radius a , aligned along the z -axis. Two distinct incident polarizations are considered with respect to the cylinder's axis: E-polarization (transverse magnetic to z , or TM) and H-polarization (transverse electric to z , or TE). For a z -invariant, normally incident wave ($\partial_z \equiv 0$) in a uniform isotropic exterior, Maxwell's curls decouple: the transverse components follow algebraically from E_z (respectively H_z), which satisfies the scalar cylindrical Helmholtz equation, while the other axial field vanishes. Outgoing separated modes therefore span the exterior field, with Fourier completeness (Appendix B.5) as the angular prerequisite. We derive both cases below.

TM (E-Polarized) Incidence

Consider a plane wave propagating in the x -direction. Using the expansion from (4.2.23), the incident field can be expressed as:

$$\vec{E}_i = E_0 e^{-ikx} \hat{z} = \hat{z} E_0 \sum_{n=-\infty}^{\infty} i^{-n} J_n(k\rho) e^{in\phi}. \quad (4.3.1)$$

We assume the scattered field takes the form:

$$\vec{E}_s = \hat{z} E_0 \sum_{n=-\infty}^{\infty} i^{-n} a_n H_n^{(2)}(k\rho) e^{in\phi}. \quad (4.3.2)$$

The Hankel function of the second kind, $H_n^{(2)}(k\rho)$, is used here to represent outgoing propagating waves. Applying the matching condition that the total tangential electric field E_z^t must be zero at the cylinder surface ($\rho = a$), we have:

$$E_z^t = E_0 \sum_{n=-\infty}^{\infty} i^{-n} [J_n(ka) + a_n H_n^{(2)}(ka)] e^{in\phi} = 0. \quad (4.3.3)$$

TM (E-Polarized) Incidence (cont.)

Because the functions $\{e^{in\phi}\}$ are mutually orthogonal on $[0, 2\pi]$, the only way for the sum in (4.3.3) to vanish for every ϕ is for each term to vanish individually. To extract the m -th term, we multiply (4.3.3) by $e^{-im\phi}$ and integrate over ϕ from 0 to 2π , using $\int_0^{2\pi} e^{i(m-n)\phi} d\phi = 2\pi\delta_{mn}$:

$$E_0 \sum_{n=-\infty}^{\infty} i^{-n} [J_n(ka) + a_n H_n^{(2)}(ka)] 2\pi\delta_{mn} = 0,$$

which collapses the sum to the single surviving term $n = m$,

$$2\pi E_0 i^{-m} [J_m(ka) + a_m H_m^{(2)}(ka)] = 0.$$

Since $E_0 \neq 0$ and $i^{-m} \neq 0$, the bracket itself must vanish for every m ,

$$J_m(ka) + a_m H_m^{(2)}(ka) = 0.$$

Relabeling $m \rightarrow n$ and solving for the coefficient then gives

$$a_n = -\frac{J_n(ka)}{H_n^{(2)}(ka)}. \quad (4.3.4)$$

TM (E-Polarized) Incidence (cont.)

Substituting this back, the scattered field is:

$$\vec{E}_s = -\hat{z}E_0 \sum_{n=-\infty}^{\infty} i^{-n} \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)}(k\rho) e^{in\phi}. \quad (4.3.5)$$

And the total field is:

$$\vec{E}_t = \hat{z}E_0 \sum_{n=-\infty}^{\infty} i^{-n} \left[J_n(k\rho) - \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)}(k\rho) \right] e^{in\phi}. \quad (4.3.6)$$

Justification of the cylindrical far-field limit. A fixed-order Hankel expansion does not alone justify taking the radial limit through an infinite modal sum. Let $k > 0$ and normalize the outgoing scalar field as $w = E_z^s/E_0$ for TM or $w = H_z^s/H_0$ for TE. On a fixed circle $\mathcal{C}_b$ with $b > a$, its boundary traces are smooth; let $\hat{n}'$ point radially outward. Impose boundedness and the uniform outgoing condition $\sqrt{\rho}(\partial_\rho + ik)w \rightarrow 0$ at infinity. For two-dimensional positions, the outgoing Green function is

$$g_2(\vec{\rho}, \vec{\rho}') = -\frac{i}{4} H_0^{(2)}(k|\vec{\rho} - \vec{\rho}'|),$$

TM (E-Polarized) Incidence (cont.)

$$(\nabla_{\perp}^2 + k^2)g_2 = -\delta^{(2)}(\vec{\rho} - \vec{\rho}').$$

Writing $s = |\vec{\rho} - \vec{\rho}'|$, the small-argument Bessel series give $g_2 = -(2\pi)^{-1} \ln s + O(1)$ and $\partial_s g_2 = -(2\pi s)^{-1} + O(s |\ln s|)$. Hence $2\pi s \partial_s g_2 \rightarrow -1$, fixing the normalization. Green's second identity on the exterior annulus gives

$$w(\vec{\rho}) = \oint_{\mathcal{C}_b} [w(\vec{\rho}') \partial_{n'} g_2 - g_2 \partial_{n'} w(\vec{\rho}')] d\ell'. \quad (4.3.7)$$

The sign follows because the annulus normal on $\mathcal{C}_b$ is $-\hat{n}'$. On an outer circle of radius L , rewrite its integrand as $w(\partial_L + ik)g_2 - g_2(\partial_L + ik)w$. The fixed-observation estimates $g_2 = O(L^{-1/2})$ and $(\partial_L + ik)g_2 = O(L^{-3/2})$, together with the conditions on w , bound that integral by $O(L^{-1/2}) + o(1)$, which vanishes.

For $\vec{\rho} = \rho \hat{\rho}(\phi)$ and $\vec{\rho}' \in \mathcal{C}_b$, $|\vec{\rho} - \vec{\rho}'| = \rho - \hat{\rho} \cdot \vec{\rho}' + O(b^2/\rho)$. The fixed-order Hankel expansions and their differentiated forms give, with $C_0 = e^{-i\pi/4}/\sqrt{8\pi k}$,

$$\begin{aligned} g_2 &= C_0 \frac{e^{-ik\rho}}{\sqrt{\rho}} e^{ik\hat{\rho} \cdot \vec{\rho}'} + O(\rho^{-3/2}), \\ \partial_{n'} g_2 &= C_0 \frac{e^{-ik\rho}}{\sqrt{\rho}} ik(\hat{n}' \cdot \hat{\rho}) e^{ik\hat{\rho} \cdot \vec{\rho}'} + O(\rho^{-3/2}). \end{aligned}$$

TM (E-Polarized) Incidence (cont.)

These estimates are uniform on the fixed circle and in ϕ ; hence

$$\begin{aligned}w(\rho, \phi) &= \frac{e^{-ik\rho}}{\sqrt{\rho}} F(\phi) + O(\rho^{-3/2}), \\F(\phi) &= C_0 \oint_{C_b} e^{ik\hat{\rho} \cdot \vec{\rho}'} [ik(\hat{n}' \cdot \hat{\rho})w - \partial_{n'} w] d\ell'.\end{aligned}\tag{4.3.8}$$

Differentiating the Green kernels before taking the limit gives the same uniform expansion after any fixed number of angular derivatives, and $\partial_\rho w = -ike^{-ik\rho} F / \sqrt{\rho} + O(\rho^{-3/2})$. The compact integral also shows that F is smooth and periodic.

For each fixed integer n , the Fourier coefficient of w satisfies the separated radial equation and outgoing condition, so $w_n(\rho) = A_n H_n^{(2)}(k\rho)$. Uniform radial convergence now permits

$$\begin{aligned}F_n &= \frac{1}{2\pi} \int_0^{2\pi} F(\phi) e^{-in\phi} d\phi \\&= \lim_{\rho \rightarrow \infty} \sqrt{\rho} e^{ik\rho} w_n(\rho) = \sqrt{\frac{2}{\pi k}} e^{i\pi/4} i^n A_n.\end{aligned}$$

TM (E-Polarized) Incidence (cont.)

Repeated angular integration by parts gives $|F_n| \leq C_p(1 + |n|)^{-p}$ for every fixed integer $p \geq 0$. Thus the Fourier series of F , and each fixed angular derivative, converges absolutely and uniformly. Reconstructing F from these coefficients justifies the modal far-field formula without assuming Hankel asymptotics uniform in n . Here $A_n = -i^{-n} J_n(ka)/H_n^{(2)}(ka)$; for the TE field derived below, $A_n = -i^{-n} J'_n(ka)/H_n^{(2)'}(ka)$. We first isolate the ρ - and n -dependence of the asymptotic Hankel function by writing (4.2.17) as

$$H_n^{(2)}(k\rho) \approx \sqrt{\frac{2}{\pi k\rho}} e^{-ik\rho} e^{in\pi/2} e^{i\pi/4},$$

where we used $e^{-i(k\rho - n\pi/2 - \pi/4)} = e^{-ik\rho} e^{in\pi/2} e^{i\pi/4}$. Substituting this into the scattered field (4.3.5) and pulling the n -independent factors out of the sum yields

$$\vec{E}_s \approx -\hat{z} E_0 \sqrt{\frac{2}{\pi k\rho}} e^{i\pi/4} e^{-ik\rho} \sum_{n=-\infty}^{\infty} i^{-n} \frac{J_n(ka)}{H_n^{(2)}(ka)} e^{in\pi/2} e^{in\phi}.$$

TM (E-Polarized) Incidence (cont.)

The n -dependent phase from the asymptotic form is $e^{in\pi/2} = i^n$, which cancels the i^{-n} already present in each term,

$$i^{-n} e^{in\pi/2} = i^{-n} i^n = 1,$$

so the summand reduces to $[J_n(ka)/H_n^{(2)}(ka)] e^{in\phi}$. Finally, the constant phase is absorbed into the radical using $e^{i\pi/4} = \sqrt{i}$, giving $\sqrt{2/(\pi k\rho)} \sqrt{i} = \sqrt{2i/(\pi k\rho)}$. Collecting the outgoing $e^{-ik\rho}/\sqrt{\rho}$ dependence, we obtain

$$\vec{E}_s \approx -\hat{z} E_0 \sqrt{\frac{2i}{\pi k}} \frac{e^{-ik\rho}}{\sqrt{\rho}} \sum_{n=-\infty}^{\infty} \frac{J_n(ka)}{H_n^{(2)}(ka)} e^{in\phi},$$

which is precisely the 2D far-field form $E_0 (e^{-ik\rho}/\sqrt{\rho}) \vec{f}(\hat{s}, \hat{i})$ of (4.1.3), with the scattering amplitude

$$\vec{f}(\hat{s}, \hat{i}) = -\hat{z} \sqrt{\frac{2i}{\pi k}} \sum_{n=-\infty}^{\infty} \frac{J_n(ka)}{H_n^{(2)}(ka)} e^{in\phi}.$$

TM (E-Polarized) Incidence (cont.)

Writing $\sqrt{i} = e^{i\pi/4}$ gives the equivalent phase-factored form $\vec{f} = -\hat{z} \sqrt{2/(\pi k)} e^{i\pi/4} \sum_{n=-\infty}^{\infty} [J_n(ka)/H_n^{(2)}(ka)] e^{in\phi}$. Since this unit-magnitude factor drops out of the modulus, the differential cross section $\sigma_d = |\vec{f}|^2$ from (4.1.5) becomes

$$\sigma_d = |\vec{f}|^2 = \frac{2}{\pi k} \left| \sum_{n=-\infty}^{\infty} \frac{J_n(ka)}{H_n^{(2)}(ka)} e^{in\phi} \right|^2.$$

Inserting this into the two-dimensional case of (4.1.6), $\sigma_{bi} = 2\pi \sigma_d$, the factor of π cancels and the prefactor reduces to $4/k$, yielding

$$\sigma_{bi} = \frac{4}{k} \left| \sum_{n=-\infty}^{\infty} \frac{J_n(ka)}{H_n^{(2)}(ka)} e^{in\phi} \right|^2. \quad (4.3.9)$$

This result is often normalized by the wavelength in the host medium, $\lambda = 2\pi/k$, which reduces the prefactor $4/k$ to

$$\frac{1}{\lambda} \cdot \frac{4}{k} = \frac{k}{2\pi} \cdot \frac{4}{k} = \frac{4}{2\pi} = \frac{2}{\pi},$$

TM (E-Polarized) Incidence (cont.)

so that dividing the echo width through by λ yields

$$\sigma_{bi}/\lambda = \frac{2}{\pi} \left| \sum_{n=-\infty}^{\infty} \frac{J_n(ka)}{H_n^{(2)}(ka)} e^{in\phi} \right|^2. \quad (4.3.10)$$

The corresponding magnetic field should not be left implicit. Faraday's law $\vec{\nabla} \times \vec{E} = -i\omega\mu\vec{H}$ and the cylindrical curl of $\hat{z}E_z$ give

$$H_\rho = \frac{i}{\omega\mu\rho} \partial_\phi E_z, \quad H_\phi = -\frac{i}{\omega\mu} \partial_\rho E_z. \quad (4.3.11)$$

Since $\omega\mu = k\eta$, applying (4.3.11) term by term to the incident field (4.3.1) yields the display below. Primes denote differentiation with respect to the complete dimensionless argument.

$$\begin{aligned} H_\rho^i &= -\frac{E_0}{\eta k \rho} \sum_{n=-\infty}^{\infty} n i^{-n} J_n(k\rho) e^{in\phi}, \\ H_\phi^i &= -\frac{iE_0}{\eta} \sum_{n=-\infty}^{\infty} i^{-n} J'_n(k\rho) e^{in\phi}. \end{aligned} \quad (4.3.12)$$

TM (E-Polarized) Incidence (cont.)

Applying the same derivatives to (4.3.5) gives the scattered magnetic field,

$$\begin{aligned} H_{\rho}^s &= \frac{E_0}{\eta k \rho} \sum_{n=-\infty}^{\infty} n i^{-n} \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)}(k\rho) e^{in\phi}, \\ H_{\phi}^s &= \frac{iE_0}{\eta} \sum_{n=-\infty}^{\infty} i^{-n} \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)'}(k\rho) e^{in\phi}. \end{aligned} \quad (4.3.13)$$

Adding (4.3.12) and (4.3.13), or directly differentiating (4.3.6), gives

$$\begin{aligned} H_{\rho}^t &= -\frac{E_0}{\eta k \rho} \sum_{n=-\infty}^{\infty} n i^{-n} \left[J_n(k\rho) - \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)}(k\rho) \right] e^{in\phi}, \\ H_{\phi}^t &= -\frac{iE_0}{\eta} \sum_{n=-\infty}^{\infty} i^{-n} \left[J_n'(k\rho) - \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)'}(k\rho) \right] e^{in\phi}. \end{aligned} \quad (4.3.14)$$

TM (E-Polarized) Incidence (cont.)

The electric surface-current density on the PEC follows from $\vec{K} = \hat{\rho} \times \vec{H}_t|_{\rho=a}$. Only H_ϕ^t contributes, so

$$\begin{aligned}\vec{K}(\phi) &= \hat{z} H_\phi^t(a, \phi) \\ &= -\hat{z} \frac{iE_0}{\eta} \sum_{n=-\infty}^{\infty} i^{-n} e^{in\phi} \\ &\quad \times \left[J'_n(ka) - \frac{J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)'}(ka) \right].\end{aligned}\tag{4.3.15}$$

The ordinary-Bessel Wronskian is

$$J_n(x) H_n^{(2)'}(x) - J_n'(x) H_n^{(2)}(x) = -\frac{2i}{\pi x}.$$

It reduces the bracket according to

$$J_n'(x) - \frac{J_n(x)}{H_n^{(2)}(x)} H_n^{(2)'}(x) = \frac{2i}{\pi x H_n^{(2)}(x)}.$$

TM (E-Polarized) Incidence (cont.)

Consequently, an equivalent compact form is

$$\vec{K}(\phi) = \hat{z} \frac{2E_0}{\pi\eta ka} \sum_{n=-\infty}^{\infty} \frac{i^{-n} e^{in\phi}}{H_n^{(2)}(ka)}. \quad (4.3.16)$$

TE (H-Polarized) Incidence

For the complementary TE case, let the incident magnetic field point along the cylinder axis and define $H_0 = E_0/\eta$. Ampère–Maxwell's law in the source-free host, $\vec{\nabla} \times \vec{H} = i\omega\epsilon\vec{E}$, gives

$$\begin{aligned}\vec{H}_i &= \hat{z}H_0e^{-ikx}, \\ \vec{E}_i &= \frac{1}{i\omega\epsilon}\vec{\nabla} \times \vec{H}_i = \hat{y}\frac{k}{\omega\epsilon}H_0e^{-ikx} = \hat{y}E_0e^{-ikx},\end{aligned}\tag{4.3.17}$$

where $k/(\omega\epsilon) = \eta$. Expanding the incident field and choosing the outgoing Hankel function for the scattered field, write

$$H_z^i = H_0 \sum_{n=-\infty}^{\infty} i^{-n} J_n(k\rho) e^{in\phi},\tag{4.3.18}$$

$$H_z^s = H_0 \sum_{n=-\infty}^{\infty} i^{-n} q_n H_n^{(2)}(k\rho) e^{in\phi}.\tag{4.3.19}$$

TE (H-Polarized) Incidence (cont.)

For a field with only a z-directed magnetic component, the cylindrical curl reduces to

$$E_\rho = \frac{1}{i\omega\epsilon\rho} \partial_\phi H_z, \quad E_\phi = -\frac{1}{i\omega\epsilon} \partial_\rho H_z. \quad (4.3.20)$$

The PEC condition is continuity to zero of the tangential electric field at $\rho = a$. Substitution of (4.3.18) and (4.3.19) into (4.3.20) gives

$$\begin{aligned} E_\phi^t(a, \phi) &= -\frac{kH_0}{i\omega\epsilon} \sum_{n=-\infty}^{\infty} i^{-n} \left[J_n'(ka) + q_n H_n^{(2)'}(ka) \right] e^{in\phi} \\ &= iE_0 \sum_{n=-\infty}^{\infty} i^{-n} \left[J_n'(ka) + q_n H_n^{(2)'}(ka) \right] e^{in\phi} = 0. \end{aligned} \quad (4.3.21)$$

Fourier orthogonality forces each bracket in (4.3.21) to vanish separately, so

$$q_n = -\frac{J_n'(ka)}{H_n^{(2)'}(ka)}. \quad (4.3.22)$$

TE (H-Polarized) Incidence (cont.)

Thus the scattered and total axial magnetic fields are

$$H_z^s = -H_0 \sum_{n=-\infty}^{\infty} i^{-n} \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)}(k\rho) e^{in\phi}, \quad (4.3.23)$$

$$H_z^t = H_0 \sum_{n=-\infty}^{\infty} i^{-n} \left[J_n(k\rho) - \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)}(k\rho) \right] e^{in\phi}. \quad (4.3.24)$$

Applying (4.3.20) term by term makes the electric fields equally explicit. In particular,

$$\begin{aligned} E_\rho^s &= -\frac{E_0}{k\rho} \sum_{n=-\infty}^{\infty} n i^{-n} \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)}(k\rho) e^{in\phi}, \\ E_\phi^s &= -iE_0 \sum_{n=-\infty}^{\infty} i^{-n} \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)'}(k\rho) e^{in\phi}, \end{aligned} \quad (4.3.25)$$

TE (H-Polarized) Incidence (cont.)

and

$$\begin{aligned} E_{\rho}^t &= \frac{E_0}{k\rho} \sum_{n=-\infty}^{\infty} n i^{-n} \left[J_n(k\rho) - \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)}(k\rho) \right] e^{in\phi}, \\ E_{\phi}^t &= iE_0 \sum_{n=-\infty}^{\infty} i^{-n} \left[J'_n(k\rho) - \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)'}(k\rho) \right] e^{in\phi}. \end{aligned} \quad (4.3.26)$$

The second bracket in (4.3.26) vanishes identically at $\rho = a$, directly checking the PEC boundary condition. The induced current is now azimuthal because

$$\begin{aligned} \vec{K}(\phi) &= \hat{\rho} \times \hat{z} H_z^t(a, \phi) = -\hat{\phi} H_z^t(a, \phi) \\ &= -\hat{\phi} H_0 \sum_{n=-\infty}^{\infty} i^{-n} \left[J_n(ka) - \frac{J'_n(ka)}{H_n^{(2)'}(ka)} H_n^{(2)}(ka) \right] e^{in\phi}. \end{aligned} \quad (4.3.27)$$

TE (H-Polarized) Incidence (cont.)

Apply (4.3.8) to $w = H_z^s/H_0$. Its differentiated limits give $\partial_\rho H_z^s = -ikH_z^s + O(\rho^{-3/2})$ and $\rho^{-1}\partial_\phi H_z^s = O(\rho^{-3/2})$. Equation (4.3.20) therefore yields $E_\phi^s = \eta H_z^s + O(\rho^{-3/2})$ and $E_\rho^s = O(\rho^{-3/2})$. Using the same phase cancellation $i^{-n}e^{in\pi/2} = 1$ gives

$$\begin{aligned}\vec{E}_s &\simeq -\hat{\phi}E_0\sqrt{\frac{2i}{\pi k}}\frac{e^{-ik\rho}}{\sqrt{\rho}}\sum_{n=-\infty}^{\infty}\frac{J'_n(ka)}{H_n^{(2)'}(ka)}e^{in\phi}, \\ \vec{f}_{\text{TE}}(\phi) &= -\hat{\phi}\sqrt{\frac{2i}{\pi k}}\sum_{n=-\infty}^{\infty}\frac{J'_n(ka)}{H_n^{(2)'}(ka)}e^{in\phi}.\end{aligned}\tag{4.3.28}$$

The incident flux may be written either as $E_0^2/(2\eta)$ or as $\eta H_0^2/2$; hence the same 2D normalization used for the TM case applies. The TE echo width and its wavelength-normalized form are consequently

$$\sigma_{bi}^{\text{TE}} = \frac{4}{k}\left|\sum_{n=-\infty}^{\infty}\frac{J'_n(ka)}{H_n^{(2)'}(ka)}e^{in\phi}\right|^2,\tag{4.3.29}$$

TE (H-Polarized) Incidence (cont.)

$$\frac{\sigma_{bi}^{\text{TE}}}{\lambda} = \frac{2}{\pi} \left| \sum_{n=-\infty}^{\infty} \frac{J'_n(ka)}{H_n^{(2)'}(ka)} e^{in\phi} \right|^2. \quad (4.3.30)$$

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Spherical Waves

This section treats wave solutions in spherical coordinates; unlike the cylindrical case, these require the special considerations developed below.

Spherical Wave Solution

The application of the vector Helmholtz equation in spherical coordinates is not straightforward, as none of the unit vectors are constant throughout space. As a result, the vector Helmholtz equation cannot be directly reduced to separate scalar Helmholtz equations for each component. To illustrate this, consider (3.1.4)

$\vec{\nabla} \times \vec{\nabla} \times \vec{A} - k^2 \vec{A} = \mu \vec{J} - i\omega\mu\epsilon \vec{\nabla} \varphi$ in spherical coordinates. By assuming that $\vec{A}$ and $\vec{J}$ have only radial component, that is

$$\vec{A} = A_r \hat{r}, \quad \vec{J} = J_r \hat{r}. \quad (4.4.1)$$

Then, the r , θ , and ϕ components of (3.1.4) are given by

$$\left[\frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 + k^2 \right] A_r = i\omega\mu\epsilon \partial_r \varphi - \mu J_r, \quad (4.4.2)$$

$$-\frac{1}{r} \partial_r \partial_\theta A_r = \frac{i\omega\mu\epsilon}{r} \partial_\theta \varphi, \quad (4.4.3)$$

$$-\frac{1}{r \sin \theta} \partial_r \partial_\phi A_r = \frac{i\omega\mu\epsilon}{r \sin \theta} \partial_\phi \varphi. \quad (4.4.4)$$

Spherical Wave Solution (cont.)

These follow from evaluating $\vec{B} = \vec{\nabla} \times \vec{A}$ for $\vec{A} = A_r \hat{r}$. With $A_\theta = A_\phi = 0$, the spherical curl gives

$$B_r = 0, \quad B_\theta = \frac{1}{r \sin \theta} \partial_\phi A_r, \quad B_\phi = -\frac{1}{r} \partial_\theta A_r.$$

Taking the curl once more, the radial component is

$$\begin{aligned} & \frac{1}{r \sin \theta} [\partial_\theta (\sin \theta B_\phi) - \partial_\phi B_\theta] \\ &= - \left[\frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 \right] A_r, \end{aligned}$$

which carries no $\partial_r^2 A_r$ term because $B_r = 0$, while the transverse components are

$$(\vec{\nabla} \times \vec{B})_\theta = \frac{1}{r} \partial_r \partial_\theta A_r, \quad (\vec{\nabla} \times \vec{B})_\phi = \frac{1}{r \sin \theta} \partial_r \partial_\phi A_r.$$

Equating $\vec{\nabla} \times \vec{\nabla} \times \vec{A} - k^2 \vec{A}$ component by component to the right-hand side $\mu \vec{J} - i\omega \mu \epsilon \vec{\nabla} \varphi$ of (3.1.4) (using $(\vec{\nabla} \varphi)_\theta = \frac{1}{r} \partial_\theta \varphi$ and $(\vec{\nabla} \varphi)_\phi = \frac{1}{r \sin \theta} \partial_\phi \varphi$) reproduces

Spherical Wave Solution (cont.)

(4.4.2)–(4.4.4). We notice that the Lorenz gauge condition (3.1.8) cannot be used to simplify (4.4.3) and (4.4.4); instead, we choose

$$\partial_r A_r + i\omega\mu\epsilon\varphi = 0. \quad (4.4.5)$$

By doing so, (4.4.3) and (4.4.4) are satisfied. Indeed, differentiating the gauge condition (4.4.5) with respect to θ and ϕ (which commute with ∂_r) gives

$$i\omega\mu\epsilon\partial_\theta\varphi = -\partial_r\partial_\theta A_r, \quad i\omega\mu\epsilon\partial_\phi\varphi = -\partial_r\partial_\phi A_r.$$

Substituting these into the right-hand sides of (4.4.3) and (4.4.4) reproduces their left-hand sides exactly, e.g.

$$\frac{i\omega\mu\epsilon}{r}\partial_\theta\varphi = -\frac{1}{r}\partial_r\partial_\theta A_r,$$

so both reduce to the identity $0 = 0$. Likewise, differentiating (4.4.5) with respect to r gives

$$i\omega\mu\epsilon\partial_r\varphi = -\partial_r^2 A_r,$$

Spherical Wave Solution (cont.)

which is the scalar-potential term on the right-hand side of (4.4.2). Moving it to the left-hand side supplies the missing radial second derivative $\partial_r^2 A_r$ and leaves only $-\mu J_r$ on the right, so that

$$\left[\partial_r^2 + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 + k^2 \right] A_r = -\mu J_r. \quad (4.4.6)$$

Let us introduce the Debye potential:

$$\pi_e = \frac{A_r}{i\omega\mu\epsilon r}. \quad (4.4.7)$$

Then (4.4.6) is transformed into the scalar Helmholtz equation in spherical coordinates with $\nabla^2 = \frac{1}{r^2} \partial_r (r^2 \partial_r) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2$. To see this, write (4.4.7) as $A_r = i\omega\mu\epsilon r \pi_e$ and substitute it into (4.4.6). Because r is constant with respect to θ and ϕ , the two transverse terms simply pass the factor $i\omega\mu\epsilon r$ through, while the radial term uses the identity

$$\partial_r^2 (r \pi_e) = r \left[\frac{1}{r^2} \partial_r (r^2 \partial_r \pi_e) \right],$$

Spherical Wave Solution (cont.)

which follows from $\partial_r^2(r\pi_e) = \partial_r(\pi_e + r\partial_r\pi_e) = 2\partial_r\pi_e + r\partial_r^2\pi_e$ and $\frac{1}{r^2}\partial_r(r^2\partial_r\pi_e) = \frac{2}{r}\partial_r\pi_e + \partial_r^2\pi_e$. Hence every term on the left of (4.4.6) carries the common factor $i\omega\mu\epsilon r$,

$$\begin{aligned} & \left[\partial_r^2 + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 + k^2 \right] A_r \\ &= i\omega\mu\epsilon r \left[\frac{1}{r^2} \partial_r (r^2 \partial_r) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) \right. \\ & \quad \left. + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 + k^2 \right] \pi_e \\ &= i\omega\mu\epsilon r (\nabla^2 + k^2) \pi_e. \end{aligned}$$

Equating this to the right-hand side $-\mu J_r$ of (4.4.6) and dividing through by $i\omega\mu\epsilon r$ yields, since $-\mu J_r/(i\omega\mu\epsilon r) = -J_r/(i\omega\epsilon r)$,

$$(\nabla^2 + k^2) \pi_e = -\frac{J_r}{i\omega\epsilon r}. \quad (4.4.8)$$

Spherical Wave Solution (cont.)

By duality, the radial magnetic current density $\vec{M} = M_r \hat{r}$ similarly gives

$$(\nabla^2 + k^2) \pi_m = -\frac{M_r}{i\omega\mu r}. \quad (4.4.9)$$

Once the solutions of (4.4.8) and (4.4.9) are found, the fields can be found by plugging (4.4.5), (4.4.7) into (3.1.3), (3.1.1) and applying the duality transform. Before these quantities are first used, define the scalar radial amplitudes and their corresponding radial vectors by

$$\Pi_e \equiv r\pi_e, \quad \vec{\Pi}_e \equiv \Pi_e \hat{r}, \quad \Pi_m \equiv r\pi_m, \quad \vec{\Pi}_m \equiv \Pi_m \hat{r}.$$

Let us first carry out the electric part explicitly. The gauge choice (4.4.5) fixes the scalar potential as $\varphi = -\frac{1}{i\omega\mu\epsilon} \partial_r A_r$, while (4.4.7) gives $A_r = i\omega\mu\epsilon r\pi_e = i\omega\mu\epsilon \Pi_e$. Substituting both into (3.1.3),

$$\begin{aligned} \vec{E} &= -\vec{\nabla}\varphi - i\omega\vec{A} \\ &= \frac{1}{i\omega\mu\epsilon} \vec{\nabla}(\partial_r A_r) - i\omega A_r \hat{r} \\ &= \vec{\nabla}(\partial_r \Pi_e) + \omega^2 \mu\epsilon \Pi_e \hat{r}, \end{aligned}$$

Spherical Wave Solution (cont.)

which, using $k^2 = \omega^2 \mu \epsilon$, is exactly the electric part $\vec{\nabla} \partial_r \Pi_e + k^2 \vec{\Pi}_e$. The magnetic contribution $-i\omega\mu \vec{\nabla} \times \vec{\Pi}_m$ is then appended by duality, yielding (4.4.12) below; the $\vec{H}$ expression follows from $\Pi_e \rightarrow \Pi_m$, $\Pi_m \rightarrow -\Pi_e$, $\mu \rightarrow \epsilon$.

To obtain the components, recall the spherical gradient

$$\vec{\nabla} f = \hat{r} \partial_r f + \hat{\theta} \frac{1}{r} \partial_\theta f + \hat{\phi} \frac{1}{r \sin \theta} \partial_\phi f,$$

and the curl of a purely radial field $\vec{\Pi}_m = \Pi_m \hat{r}$,

$$\vec{\nabla} \times \vec{\Pi}_m = \hat{\theta} \frac{1}{r \sin \theta} \partial_\phi \Pi_m - \hat{\phi} \frac{1}{r} \partial_\theta \Pi_m.$$

Applying the gradient to the scalar $\partial_r \Pi_e$ and adding the $k^2 \Pi_e \hat{r}$ term gives the electric part of each component, while $-i\omega\mu \vec{\nabla} \times \vec{\Pi}_m$ supplies the magnetic part; the $\hat{r}$ -curl vanishes, so E_r has no magnetic contribution.

$$\vec{E} = \vec{\nabla} \partial_r \Pi_e + k^2 \vec{\Pi}_e - i\omega\mu \vec{\nabla} \times \vec{\Pi}_m, \quad (4.4.10)$$

$$\vec{H} = \vec{\nabla} \partial_r \Pi_m + k^2 \vec{\Pi}_m + i\omega\epsilon \vec{\nabla} \times \vec{\Pi}_e. \quad (4.4.11)$$

Spherical Wave Solution (cont.)

The source-free Maxwell curls provide an independent sign check on this reconstruction. If $\pi = \Pi/r$ obeys $(\nabla^2 + k^2)\pi = 0$, multiplication by r gives

$$\partial_r^2 \Pi + \frac{1}{r^2} \nabla_a^2 \Pi + k^2 \Pi = 0,$$

where $\vec{\nabla}_a \equiv \hat{\theta} \partial_\theta + \hat{\phi} (\sin \theta)^{-1} \partial_\phi$ is the dimensionless angular gradient on the unit sphere, so that $\vec{\nabla} = \hat{r} \partial_r + r^{-1} \vec{\nabla}_a$, and $\nabla_a^2 = \partial_\theta^2 + \cot \theta \partial_\theta + (\sin \theta)^{-2} \partial_\phi^2$ is the scalar angular Laplacian with $\nabla_a^2 Y_{nm} = -n(n+1) Y_{nm}$ (proved in Appendix C.6). The curl-curl components derived above then give

$$\vec{\nabla} \times \vec{\nabla} \times (\Pi \hat{r}) = \vec{\nabla} \partial_r \Pi + k^2 \Pi \hat{r}.$$

Thus the electric-potential fields may be written $\vec{E}_e = \vec{\nabla} \times \vec{\nabla} \times \vec{\Pi}_e$ and $\vec{H}_e = i\omega\epsilon \vec{\nabla} \times \vec{\Pi}_e$. Their curls are

$$\vec{\nabla} \times \vec{E}_e = k^2 \vec{\nabla} \times \vec{\Pi}_e = -i\omega\mu \vec{H}_e,$$

$$\vec{\nabla} \times \vec{H}_e = i\omega\epsilon \vec{\nabla} \times \vec{\nabla} \times \vec{\Pi}_e = i\omega\epsilon \vec{E}_e,$$

Spherical Wave Solution (cont.)

because $k^2 = \omega^2 \mu \epsilon$. For the magnetic potential, $\vec{H}_m = \vec{\nabla} \times \vec{\nabla} \times \vec{\Pi}_m$ and $\vec{E}_m = -i\omega\mu\vec{\nabla} \times \vec{\Pi}_m$ give the same two Maxwell equations. This verifies both curl-term signs in the combined fields above.

In terms of spherical components, the electric field is expressed as:

$$E_r = (\partial_r^2 + k^2) \Pi_e, \quad (4.4.12a)$$

$$E_\theta = \frac{1}{r} \partial_r \partial_\theta \Pi_e - \frac{i\omega\mu}{r \sin \theta} \partial_\phi \Pi_m, \quad (4.4.12b)$$

$$E_\phi = \frac{1}{r \sin \theta} \partial_r \partial_\phi \Pi_e + \frac{i\omega\mu}{r} \partial_\theta \Pi_m, \quad (4.4.12c)$$

and the magnetic field is expressed as:

$$H_r = (\partial_r^2 + k^2) \Pi_m, \quad (4.4.13a)$$

$$H_\theta = \frac{1}{r} \partial_r \partial_\theta \Pi_m + \frac{i\omega\epsilon}{r \sin \theta} \partial_\phi \Pi_e, \quad (4.4.13b)$$

$$H_\phi = \frac{1}{r \sin \theta} \partial_r \partial_\phi \Pi_m - \frac{i\omega\epsilon}{r} \partial_\theta \Pi_e. \quad (4.4.13c)$$

Spherical Wave Solution (cont.)

We derived the source terms in (4.4.8)–(4.4.9) only for radial electric and magnetic currents. Transverse source components require a separate projection and cannot simply be inserted into those right-hand sides. In any homogeneous region away from all sources, the right-hand sides are set to zero.

The scalar basis Y_{nm} is normalized and proved complete in Appendix C.6.1. Write $\lambda_n = n(n+1)$ and, for $n \geq 1$, define

$$\vec{U}_{nm} = \vec{\nabla}_a Y_{nm}, \quad \vec{V}_{nm} = \hat{r} \times \vec{\nabla}_a Y_{nm}.$$

We first prove tangential completeness. For a tangential field $\vec{F}$, define its scalar surface divergence and curl by

$$\begin{aligned} \text{div}_a \vec{F} &= \frac{\partial_\theta(\sin \theta F_\theta) + \partial_\phi F_\phi}{\sin \theta}, \\ \text{curl}_a \vec{F} &= \frac{\partial_\theta(\sin \theta F_\phi) - \partial_\phi F_\theta}{\sin \theta}. \end{aligned}$$

Direct substitution, followed by integration by parts on the closed sphere, gives

$$\begin{aligned} \text{div}_a \vec{U}_{nm} &= -\lambda_n Y_{nm}, & \text{curl}_a \vec{U}_{nm} &= 0, \\ \text{div}_a \vec{V}_{nm} &= 0, & \text{curl}_a \vec{V}_{nm} &= -\lambda_n Y_{nm}, \end{aligned}$$

Spherical Wave Solution (cont.)

and

$$\int \vec{U}_{nm}^* \cdot \vec{U}_{pq} d\Omega = \lambda_n \delta_{np} \delta_{mq},$$

$$\int \vec{V}_{nm}^* \cdot \vec{V}_{pq} d\Omega = \lambda_n \delta_{np} \delta_{mq},$$

$$\int \vec{U}_{nm}^* \cdot \vec{V}_{pq} d\Omega = 0.$$

For example, the first integral is $-\int Y_{nm}^* \nabla_a^2 Y_{pq} d\Omega$; the mixed one is zero because $\text{div}_a \vec{V}_{pq} = 0$.

For a smooth tangential $\vec{F}$, put $d = \text{div}_a \vec{F}$ and $c = \text{curl}_a \vec{F}$. Both have zero mean by the divergence and Stokes theorems on the closed sphere. Expand them in the scalar basis and set

$$\Phi = - \sum_{n \geq 1} \sum_{m=-n}^n \frac{d_{nm}}{\lambda_n} Y_{nm}, \quad \Psi = - \sum_{n \geq 1} \sum_{m=-n}^n \frac{c_{nm}}{\lambda_n} Y_{nm}.$$

The coefficient and derivative bounds in Appendix C.6.1 justify these differentiated series. Thus $\vec{F}_0 = \vec{\nabla}_a \Phi + \hat{r} \times \vec{\nabla}_a \Psi$ has the same divergence and curl as $\vec{F}$. Their

Spherical Wave Solution (cont.)

difference $\vec{R}$ has zero surface curl. Every closed curve on S^2 bounds a surface, so Stokes' theorem makes its line integral zero; path integration gives a single-valued h with $\vec{R} = \vec{\nabla}_a h$. Zero divergence gives $\nabla_a^2 h = 0$, and

$$\int |\vec{R}|^2 d\Omega = - \int h^* \nabla_a^2 h d\Omega = 0.$$

Therefore $\vec{F} = \vec{F}_0$. Smooth tangential fields are dense in L^2 (approximate components in regular charts and project tangentially), so the two tangential families are complete there as well. Together with $Y_{nm}\hat{r}$ they span $L^2(S^2)^3$. Dividing each tangential member by $\sqrt{\lambda_n}$ makes the vector family orthonormal.

Angular completeness must still be connected to Maxwell's equations. For one n, m , expand

$$\vec{E} = e_r Y_{nm} \hat{r} + e_g \vec{U}_{nm} + e_c \vec{V}_{nm},$$

$$\vec{H} = h_r Y_{nm} \hat{r} + h_g \vec{U}_{nm} + h_c \vec{V}_{nm},$$

Spherical Wave Solution (cont.)

where all six coefficients depend on r . The spherical divergence and curl formulas, applied to a generic coefficient triple (f_r, f_g, f_c) , give

$$\vec{\nabla} \cdot \vec{F} = \left[\frac{(r^2 f_r)'}{r^2} - \frac{\lambda_n f_g}{r} \right] Y_{nm},$$
$$\vec{\nabla} \times \vec{F} = -\frac{\lambda_n f_c}{r} Y_{nm} \hat{r} - \frac{(r f_c)'}{r} \vec{U}_{nm} + \frac{(r f_g)' - f_r}{r} \vec{V}_{nm}.$$

Let $p_e = r^2 e_r / \lambda_n$ and $p_m = r^2 h_r / \lambda_n$. The two divergence equations and the radial parts of the two Maxwell curl equations imply

$$e_g = p_e' / r, \quad h_g = p_m' / r,$$
$$e_c = i\omega\mu p_m / r, \quad h_c = -i\omega\epsilon p_e / r.$$

For example, the radial part of $\vec{\nabla} \times \vec{E} = -i\omega\mu \vec{H}$ is $-\lambda_n e_c / r = -i\omega\mu \lambda_n p_m / r^2$. The remaining $\vec{V}_{nm}$ components give, separately for p_e, p_m ,

$$p'' + \left[k^2 - \frac{\lambda_n}{r^2} \right] p = 0.$$

Spherical Wave Solution (cont.)

The $\vec{U}_{nm}$ curl components are then satisfied by the differentiated relations above. Setting $\Pi_e = p_e Y_{nm}$ and $\Pi_m = p_m Y_{nm}$ reproduces (4.4.12) and (4.4.13) component by component. Thus every sufficiently regular source-free field in a homogeneous spherical ball or shell is generated by the two Debye potentials, with the radial solutions allowed by regularity and boundary conditions. This establishes both the angular span and the two physical polarizations, rather than inferring completeness from orthogonality. There is no missing dynamic $n = 0$ mode. For $n = 0$ the angular derivatives vanish, while the radial equation gives $(\partial_r^2 + k^2)[rR_0(kr)] = 0$; hence every field component reconstructed above is zero. Equivalently, a nonzero time-harmonic electromagnetic monopole would contradict the source-free Maxwell curl equations. Thus the physical sum begins at $n = 1$, and the two scalar Debye potentials form a complete representation under the regularity and spherical-region hypotheses just stated. When $\Pi_m = 0$ only the TM family remains; when $\Pi_e = 0$ only the TE family remains.

Now consider the scalar Helmholtz equation in spherical coordinates

$$\frac{1}{r^2} \partial_r(r^2 \partial_r \psi) + \frac{1}{r^2 \sin \theta} \partial_\theta(\sin \theta \partial_\theta \psi) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 \psi + k^2 \psi = 0. \quad (4.4.14)$$

Spherical Wave Solution (cont.)

Using separation of variables, let

$$\psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi). \quad (4.4.15)$$

Plug into (4.4.14), multiply $r^2 \sin^2 \theta$ and divide it by ψ , we get

$$\begin{aligned} \frac{\sin^2 \theta}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} \\ + (kr \sin \theta)^2 = 0. \end{aligned} \quad (4.4.16)$$

Similar to the cylindrical case, for the Φ term, we have

$$\frac{d^2 \Phi}{d\phi^2} + \nu^2 \Phi = 0, \quad \Phi = e^{\pm i\nu\phi}, \quad (4.4.17)$$

where ν is a constant. Plug into (4.4.16) and divide it by $\sin^2 \theta$, we get

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) - \frac{\nu^2}{\sin^2 \theta} + (kr)^2 = 0. \quad (4.4.18)$$

Spherical Wave Solution (cont.)

Now, introduce a dimensionless separation constant λ (distinct from the host-medium wavelength $\lambda = 2\pi/k$ used earlier) via

$$\frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) - \frac{\nu^2}{\sin^2 \theta} = -\lambda, \quad (4.4.19)$$

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + (kr)^2 = \lambda. \quad (4.4.20)$$

Since Φ is periodic over 2π , we let $\nu = m$ an integer value, and from (4.4.19), we get

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \left(\lambda - \frac{m^2}{\sin^2 \theta} \right) \Theta = 0. \quad (4.4.21)$$

Let us consider two cases: the solution is azimuthal symmetric or non-azimuthal symmetric. If it is azimuthal symmetric, we have $m = 0$ and (4.4.21) is reduced to

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \lambda \Theta = 0. \quad (4.4.22)$$

For a nonzero angular eigenfunction to remain finite at both poles, the dimensionless separation eigenvalue must be $\lambda = n(n+1)$, where n is a nonnegative integer. This

Spherical Wave Solution (cont.)

restriction is not arbitrary. Introducing the Legendre argument $x = \cos \theta$ (distinct from the Mie size parameter $x = ka$ used later), so that $\sin^2 \theta = 1 - x^2$ and $\sin \theta d\theta = -dx$, the azimuthally symmetric equation (4.4.22) takes the self-adjoint form

$$\frac{d}{dx} \left[(1 - x^2) \frac{d\Theta}{dx} \right] + \lambda \Theta = 0,$$

valid on $-1 \leq x \leq 1$. The coefficient $1 - x^2$ vanishes at $x = \pm 1$ (i.e. at $\cos \theta = \pm 1$), so these endpoints are regular singular points. The point $x = 0$ is ordinary; seeking a Taylor series $\Theta = \sum_{q \geq 0} a_q x^q$ there and matching like powers gives the two-term recurrence

$$a_{q+2} = \frac{q(q+1) - \lambda}{(q+1)(q+2)} a_q.$$

The recurrence separates into even and odd coefficient chains. For generic λ they do not terminate; a suitable linear combination can be regular at one endpoint, but the second endpoint then contains the logarithmically singular Legendre solution. Regularity at both

Spherical Wave Solution (cont.)

endpoints is obtained when one chain terminates, which requires its numerator to vanish at some integer $q = n$:

$$\lambda = n(n+1), \quad n = 0, 1, 2, \dots,$$

in which case the series collapses to the Legendre polynomial $P_n(x) = P_n(\cos \theta)$ and the second, unbounded solution is discarded. Thus, we arrive at the following equation

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + n(n+1)\Theta = 0, \quad (4.4.23)$$

which is known as the Legendre equation, and their solutions are known as the Legendre polynomials:

$$P_n(\cos \theta) = \frac{1}{2^n n!} \left(\frac{d}{d \cos \theta} \right)^n (\cos^2 \theta - 1)^n. \quad (4.4.24)$$

The Legendre polynomials form a complete orthogonal basis of $L^2([-1, 1])$. Their orthogonality relation is

$$\int_{-1}^1 P_n(\cos \theta) P_{n'}(\cos \theta) d \cos \theta = \frac{2}{2n+1} \delta_{nn'}. \quad (4.4.25)$$

Spherical Wave Solution (cont.)

Hence, if $g(x) = f(\arccos x)$ belongs to $L^2([-1, 1])$, its Legendre series is

$$f(\theta) = \sum_{n=0}^{\infty} a_n P_n(\cos \theta) \quad \text{in } L^2([0, \pi], \sin \theta d\theta), \quad (4.4.26)$$

where

$$a_n = \frac{2n+1}{2} \int_0^\pi f(\theta) P_n(\cos \theta) \sin \theta d\theta. \quad (4.4.27)$$

The L^2 statement does not by itself assert pointwise convergence; such a claim requires additional regularity of f and must be checked under the appropriate Legendre-series convergence theorem. On the other hand, if the solution is not azimuthally symmetric, we obtain the associated Legendre equation

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \left[n(n+1) - \frac{m^2}{\sin^2 \theta} \right] \Theta = 0. \quad (4.4.28)$$

The solutions to (4.4.28) are the associated Legendre functions of the first kind $P_n^m(\cos \theta)$ and second kind $Q_n^m(\cos \theta)$. Here, m is the integer azimuthal order introduced above. Requiring one angular eigenfunction to remain bounded at *both* poles selects

Spherical Wave Solution (cont.)

integer $n \geq 0$ and integer m with $|m| \leq n$; the bounded first-kind solution is $P_n^m(\cos \theta)$, while the second-kind solution is singular at a pole. The real azimuthal basis has only the constant cosine mode $\cos 0\phi = 1$ at $m = 0$; for every $m = 1, \dots, n$ it has the independent pair $\cos m\phi$ and $\sin m\phi$. Throughout this book we use the phase-free engineering convention $P_n^m(\cos \theta) = \sin^m \theta P_n^{(m)}(\cos \theta)$, $0 \leq m \leq n$, where $P_n^{(m)}$ denotes the m^{th} derivative of the n^{th} -order Legendre polynomial with respect to $\cos \theta$. It omits the Condon–Shortley factor $(-1)^m$; the conversion to sources that include that phase is given with (C.3.1).

Now let us consider (4.4.20). Substituting $\lambda = n(n+1)$ and letting

$$R = R_n(kr) = \sqrt{\frac{\pi}{2kr}} B_{n+1/2}(kr) = \sqrt{\frac{\pi}{2kr}} B, \quad (4.4.29)$$

we first note that (4.4.20) may be written in the product form

$$\frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + [(kr)^2 - n(n+1)] R = 0.$$

Spherical Wave Solution (cont.)

Writing $R = \sqrt{\pi/2k} r^{-1/2} B$ from (4.4.29), the radial derivative is

$$r^2 \frac{dR}{dr} = \sqrt{\frac{\pi}{2k}} \left[r^{3/2} \frac{dB}{dr} - \frac{1}{2} r^{1/2} B \right],$$

so that

$$\frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = \sqrt{\frac{\pi}{2k}} \left[r^{3/2} \frac{d^2 B}{dr^2} + r^{1/2} \frac{dB}{dr} - \frac{1}{4} r^{-1/2} B \right].$$

Adding $[(kr)^2 - n(n+1)]R = \sqrt{\pi/2k} r^{-1/2} [(kr)^2 - n(n+1)]B$ and multiplying through by $r^{-3/2} \sqrt{2k/\pi}$ gives

$$\frac{d^2 B}{dr^2} + \frac{1}{r} \frac{dB}{dr} + \left[k^2 - \frac{n(n+1) + \frac{1}{4}}{r^2} \right] B = 0.$$

Spherical Wave Solution (cont.)

Since $n(n+1) + \frac{1}{4} = (n + \frac{1}{2})^2$, this is

$$\frac{d^2 B}{dr^2} + \frac{1}{r} \frac{dB}{dr} + \left[k^2 - \frac{(n + 1/2)^2}{r^2} \right] B = 0, \quad (4.4.30)$$

which is the Bessel equation and $B_{n+1/2}$ is the Bessel function of order $n + 1/2$. (4.4.29) is known as the spherical Bessel function. We often use lowercase letters to denote the spherical forms of the Bessel and Hankel functions. For example, the spherical Bessel and Hankel functions are defined as

$$j_n(kr) = \sqrt{\frac{\pi}{2kr}} J_{n+1/2}(kr), \quad h_n^{(2)}(kr) = \sqrt{\frac{\pi}{2kr}} H_{n+1/2}^{(2)}(kr).$$

Spherical Wave Transformation

Similar to (4.2.18), consider an x-polarized plane wave propagating in the +z-direction:

$$\vec{E} = \hat{x} E_0 e^{-ikz} = \hat{x} E_0 e^{-ikr \cos \theta}. \quad (4.4.31)$$

Let us expand the exponential term into an infinite series of spherical waves. The *scalar phase factor* $e^{-ikr \cos \theta}$ is azimuthally symmetric, so its scalar expansion has $m = 0$; the fixed Cartesian polarization vector $\hat{x}$ itself is not azimuthally symmetric when resolved into spherical components. Furthermore, because the solution must remain finite at the origin, we have

$$e^{-ikr \cos \theta} = \sum_{n=0}^{\infty} p_n^{(\text{sph})} j_n(kr) P_n(\cos \theta). \quad (4.4.32)$$

The superscript distinguishes these temporary scalar plane-wave coefficients from the Mie scattering coefficients introduced below. To find $p_n^{(\text{sph})}$, use the orthogonality relation (4.4.25):

$$p_n^{(\text{sph})} j_n(kr) = \frac{2n+1}{2} \int_{-1}^1 e^{-ikr \cos \theta} P_n(\cos \theta) d \cos \theta. \quad (4.4.33)$$

Spherical Wave Transformation (cont.)

The required integral identity can be derived without skipping the projection step. Put $u = \cos \theta$ and $z = kr$. Rodrigues' formula and n integrations by parts give

$$\begin{aligned} I_n(z) &= \int_{-1}^1 e^{-izu} P_n(u) du \\ &= \frac{1}{2^n n!} \int_{-1}^1 e^{-izu} \frac{d^n}{du^n} (u^2 - 1)^n du \\ &= \frac{(-i)^n z^n}{2^n n!} \int_{-1}^1 e^{-izu} (1 - u^2)^n du. \end{aligned}$$

All boundary terms vanish because $(u^2 - 1)^n$ and its first $n - 1$ derivatives vanish at $u = \pm 1$. The integral representation

$$j_n(z) = \frac{z^n}{2^{n+1} n!} \int_{-1}^1 e^{-izu} (1 - u^2)^n du$$

then yields

$$\int_{-1}^1 e^{-ikr \cos \theta} P_n(\cos \theta) d \cos \theta = 2i^{-n} j_n(kr), \quad (4.4.34)$$

Spherical Wave Transformation (cont.)

because $(-i)^n = i^{-n}$. For fixed n , $j_n(kr)$ is not identically zero, so the same cancellation argument used above gives a coefficient independent of r :

$$p_n^{(\text{sph})} = (2n+1)i^{-n}. \quad (4.4.35)$$

Thus, we have the following expansion

$$\begin{aligned} E_x &= E_0 e^{-ikz} = E_0 e^{-ikr \cos \theta} \\ &= E_0 \sum_{n=0}^{\infty} (2n+1) i^{-n} j_n(kr) P_n(\cos \theta). \end{aligned} \quad (4.4.36)$$

This is called the spherical wave transformation which expands a plane wave to a sum of spherical waves.

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Scattering from Dielectric Spheres

The exact solution for the scattering of a plane electromagnetic wave by an isotropic, homogeneous dielectric sphere of arbitrary size is commonly known as Mie theory.

Consider a dielectric sphere of radius a , permittivity ϵ_d , and permeability μ_d , placed in a host medium with permittivity ϵ and permeability μ . We use

$$\begin{aligned} k &= \omega\sqrt{\epsilon\mu}, & \eta &= \frac{\omega\mu}{k} = \frac{k}{\omega\epsilon}, \\ k_d &= \omega\sqrt{\epsilon_d\mu_d}, & \eta_d &= \frac{\omega\mu_d}{k_d} = \frac{k_d}{\omega\epsilon_d}. \end{aligned} \tag{4.5.1}$$

For the lossless host, k and η are chosen positive. All interior square roots are taken consistently with k_d ; with the $e^{i\omega t}$ convention, a passive lossy material has the branch $\text{Im } k_d \leq 0$. The normalization and divided coefficient formulas below assume $\epsilon_d \neq 0$, $\mu_d \neq 0$, and hence $k_d \neq 0$. The exact zero-index cases $\epsilon_d = 0$ or $\mu_d = 0$ must instead be obtained from the undivided tangential-field matching equations by taking the appropriate material limit; direct substitution into expressions containing $1/k_d$ or $1/\eta_d$ is not defined there. Nonzero epsilon-near-zero and mu-near-zero parameters remain

Scattering from Dielectric Spheres (cont.)

algebraically valid inputs to the same formulas, although numerically stable evaluation may require rescaling quantities that contain the small divisors $1/k_d$ or $1/\eta_d$. Let the incident electric field be an x-polarized plane wave propagating in the +z-direction. The radial component of the incident electric field can be expressed as

$$\begin{aligned} E_r^i &= E_0 e^{-ikz} \hat{x} \cdot \hat{r} = E_0 \sin \theta \cos \phi e^{-ikr \cos \theta} \\ &= \frac{1}{ikr} E_0 \cos \phi \partial_\theta e^{-ikr \cos \theta}. \end{aligned} \quad (4.5.2)$$

Expanding the phase factor via (4.4.36) then gives

$$\begin{aligned} E_r^i &= \frac{E_0 \cos \phi}{ikr} \partial_\theta \sum_{n=0}^{\infty} (2n+1) i^{-n} j_n(kr) P_n(\cos \theta) \\ &= \frac{iE_0 \cos \phi}{(kr)^2} \sum_{n=1}^{\infty} (2n+1) i^{-n} \hat{j}_n(kr) P_n^1(\cos \theta), \end{aligned} \quad (4.5.3)$$

Scattering from Dielectric Spheres (cont.)

where

$$-\partial_{\theta} P_n(\cos \theta) = P_n^1(\cos \theta), \quad (4.5.4)$$

$$\hat{j}_n(kr) = krj_n(kr). \quad (4.5.5)$$

To find the electric Debye potential of the incident field, (4.4.12a) must be satisfied:

$$E_r^i = (\partial_r^2 + k^2) \Pi_e^i = (\partial_r^2 + k^2) (r\pi_e^i). \quad (4.5.6)$$

Let us expand the Debye potential in terms of spherical harmonics

$$\pi_e^i = \sum_{n=0}^{\infty} \sum_{m=0}^n j_n(kr) P_n^m(\cos \theta) (A_{mn} \cos m\phi + B_{mn} \sin m\phi). \quad (4.5.7)$$

Scattering from Dielectric Spheres (cont.)

Here and in every uniform real-basis double sum, $B_{0n} \equiv 0$ rather than an independent unknown: the $m = 0$ contribution is solely $A_{0n}P_n(\cos \theta)$ because $\sin 0\phi = 0$, so the sine-cosine pair begins at $m = 1$. Substituting (4.5.7) into (4.5.6), we get

$$E_r^i = \sum_{n=0}^{\infty} \sum_{m=0}^n \frac{n(n+1)}{kr^2} \hat{j}_n(kr) P_n^m(\cos \theta) \quad (4.5.8)$$
$$(A_{mn} \cos m\phi + B_{mn} \sin m\phi).$$

Here, the standard identity used in the derivation of (4.5.8) is established as follows: since the operator $\partial_r^2 + k^2$ in (4.5.6) acts only on the radial factor, applying it term by term to (4.5.7) reduces the problem to evaluating $(\partial_r^2 + k^2)[r j_n(kr)]$. With $\lambda = n(n+1)$ and $R = j_n(kr)$, the radial equation (4.4.20) reads

$$\frac{d}{dr} \left(r^2 \frac{dj_n}{dr} \right) + [(kr)^2 - n(n+1)] j_n = 0,$$

Scattering from Dielectric Spheres (cont.)

which, after expanding the first term and dividing by r , becomes

$$rk^2 j_n''(kr) + 2kj_n'(kr) + \left[k^2 r - \frac{n(n+1)}{r} \right] j_n(kr) = 0,$$

On the other hand, differentiating $r j_n(kr)$ twice gives

$$\begin{aligned} (\partial_r^2 + k^2)[r j_n(kr)] &= 2kj_n'(kr) + rk^2 j_n''(kr) + k^2 r j_n(kr) \\ &= \frac{n(n+1)}{r} j_n(kr), \end{aligned}$$

where the last equality follows directly from (4.4.20). For $n = 0$ the prefactor $n(n+1)$ in (4.5.8) vanishes, so the $n = 0$ Debye potential generates no electromagnetic field; A_{00} is therefore set to zero as a normalization choice. For every physical mode $n \geq 1$, comparing (4.5.8) and (4.5.3), the right-hand side of (4.5.3) carries the single azimuthal harmonic $\cos \phi$ and the single associated-Legendre order P_n^1 , whereas (4.5.8) contains the full real basis $\{1\} \cup \{\cos m\phi, \sin m\phi\}_{m \geq 1}$. Fourier orthogonality over $[0, 2\pi]$ selects

Scattering from Dielectric Spheres (cont.)

the $m = 1$ cosine harmonic, and fixed- m associated-Legendre orthogonality separates n . Hence, for $n \geq 1$, only the $m = 1$ cosine term survives,

$$A_{mn} = 0 \quad (m \neq 1), \quad B_{mn} = 0.$$

Equating the remaining n -th coefficients of $\cos \phi P_n^1 \hat{j}_n(kr)$ in (4.5.8) and (4.5.3) gives

$$\frac{n(n+1)}{kr^2} A_{1n} = \frac{i(2n+1) i^{-n}}{(kr)^2} E_0.$$

By the same cancellation argument used above— $\hat{j}_n(kr)$, $\cos \phi$, and P_n^1 are nonzero away from isolated points—the common factors, including $1/r^2$, cancel to give a coefficient identity independent of r . One extra power of k remains on the right, so

$$A_{1n} = E_0 i^{1-n} \frac{2n+1}{k n(n+1)}.$$

Scattering from Dielectric Spheres (cont.)

Writing the phase as $i^{1-n} = (i^{-1})^{n-1} = (-i)^{n-1}$, this is

$$\begin{cases} A_{mn} = 0, & m \neq 1, \\ A_{1n} = E_0(-i)^{n-1} \frac{2n+1}{kn(n+1)}, \end{cases} \quad (4.5.9)$$

$$B_{mn} = 0. \quad (4.5.10)$$

Thus, we have

$$\pi_e^i = \frac{E_0 \cos \phi}{k^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} \hat{j}_n(kr) P_n^1(\cos \theta). \quad (4.5.11)$$

The magnetic potential follows without skipping the corresponding projection. Faraday's law for the incident plane wave gives

$$\vec{H}_i = \frac{1}{\eta} \hat{z} \times \vec{E}_i = \hat{y} \frac{E_0}{\eta} e^{-ikz},$$

Scattering from Dielectric Spheres (cont.)

and hence

$$\begin{aligned} H_r^i &= \vec{H}_i \cdot \hat{r} = \frac{E_0}{\eta} \sin \theta \sin \phi e^{-ikr \cos \theta} \\ &= \frac{E_0 \sin \phi}{i\eta kr} \partial_\theta e^{-ikr \cos \theta}. \end{aligned}$$

Using the scalar plane-wave expansion (4.4.36), the phase-free identity $-\partial_\theta P_n(\cos \theta) = P_n^1(\cos \theta)$, and $j_n(kr) = \hat{j}_n(kr)/(kr)$ gives

$$H_r^i = \frac{iE_0 \sin \phi}{\eta(kr)^2} \sum_{n=1}^{\infty} (2n+1) i^{-n} \hat{j}_n(kr) P_n^1(\cos \theta).$$

Equating this term by term to $H_r^i = (\partial_r^2 + k^2)(r\pi_m^i)$ repeats the radial identity (4.5.8) with $E_0 \cos \phi$ replaced by $E_0 \sin \phi / \eta$. Therefore

$$\pi_m^i = \frac{E_0 \sin \phi}{\eta k^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} \hat{j}_n(kr) P_n^1(\cos \theta). \quad (4.5.12)$$

Scattering from Dielectric Spheres (cont.)

To find the scattered field inside and outside the sphere, let us first specify the matching conditions to be applied at $r = a$:

$$E_\theta = E_\theta^d, \quad E_\phi = E_\phi^d, \quad (4.5.13)$$

$$H_\theta = H_\theta^d, \quad H_\phi = H_\phi^d, \quad (4.5.14)$$

where the superscript d denotes the field inside the sphere. From (4.4.12) and (4.4.13), the matching conditions involve both $\pi_e = \pi_e^i + \pi_e^s$ and $\pi_m = \pi_m^i + \pi_m^s$, making them coupled. To simplify the analysis, it is convenient to decouple them by deriving matching conditions for π_e and π_m individually. To achieve this, we consider a linear combination of E_θ and E_ϕ chosen so that all terms involving π_m cancel; applying a similar technique to the magnetic components then gives, at $r = a$:

$$\frac{1}{r} \partial_r (r \pi_e) = \frac{1}{r} \partial_r (r \pi_e^d), \quad \epsilon \pi_e = \epsilon_d \pi_e^d, \quad (4.5.15)$$

$$\frac{1}{r} \partial_r (r \pi_m) = \frac{1}{r} \partial_r (r \pi_m^d), \quad \mu \pi_m = \mu_d \pi_m^d. \quad (4.5.16)$$

Scattering from Dielectric Spheres (cont.)

To justify these conditions, substitute the field components (4.4.12) into the two independent angular combinations of the tangential field. Writing $\Pi_e = r\pi_e$ and $\Pi_m = r\pi_m$, the divergence-type combination of E_θ and E_ϕ gives

$$\begin{aligned}\partial_\theta(\sin\theta E_\theta) + \partial_\phi E_\phi \\&= \frac{1}{r} \partial_r \left[\partial_\theta(\sin\theta \partial_\theta \Pi_e) + \frac{1}{\sin\theta} \partial_\phi^2 \Pi_e \right] \\&\quad - i\omega\mu \frac{1}{r} (\partial_\theta \partial_\phi \Pi_m - \partial_\phi \partial_\theta \Pi_m).\end{aligned}$$

The Π_m contribution vanishes by the equality of mixed partial derivatives, so all magnetic terms cancel. Because the angular operator commutes with ∂_r and with multiplication by r , the surviving electric part is that same operator acting on $\frac{1}{r} \partial_r(r\pi_e)$,

$$\partial_\theta(\sin\theta E_\theta) + \partial_\phi E_\phi = \left[\partial_\theta(\sin\theta \partial_\theta) + \frac{1}{\sin\theta} \partial_\phi^2 \right] \frac{1}{r} \partial_r(r\pi_e).$$

Scattering from Dielectric Spheres (cont.)

The bracketed operator equals $\sin \theta \nabla_a^2$, where ∇_a^2 is the angular Laplacian defined explicitly below. Acting on the *full* angular mode $P_n^1(\cos \theta) \cos \phi$ (or its sine counterpart) it gives $-n(n+1) \sin \theta$ times that mode, by (4.4.28). This common angular eigenfactor is the same inside and outside the sphere. Completeness and orthogonality of the angular modes therefore force continuity of each radial coefficient, giving the first condition in (4.5.15),

$$\frac{1}{r} \partial_r (r \pi_e) \Big|_{r=a} = \frac{1}{r} \partial_r (r \pi_e^d) \Big|_{r=a}.$$

The complementary, curl-type combination of H_θ and H_ϕ from (4.4.13) cancels Π_m in the same way and isolates Π_e with the prefactor $i\omega\epsilon$,

$$\begin{aligned} \frac{1}{\sin \theta} [\partial_\theta (\sin \theta H_\phi) - \partial_\phi H_\theta] \\ = - \frac{i\omega\epsilon}{r} \nabla_a^2 (r \pi_e), \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

where $\nabla_a^2 = \partial_\theta^2 + \cot \theta \partial_\theta + \frac{1}{\sin^2 \theta} \partial_\phi^2$ is the angular Laplacian. Continuity of the tangential field again removes the common eigenvalue $-n(n+1)$ of the full angular mode $P_n^1(\cos \theta) \cos \phi$ (or its sine counterpart), leaving the algebraic condition

$$\epsilon \pi_e = \epsilon_d \pi_e^d \quad \text{at } r = a.$$

By duality $(\vec{E}, \vec{H}, \epsilon, \mu) \rightarrow (\vec{H}, -\vec{E}, \mu, \epsilon)$ the magnetic potential obeys the same pair of relations. The divergence-type combination $\partial_\theta(\sin \theta H_\theta) + \partial_\phi H_\phi$ supplies the radial-derivative continuity, while the curl-type combination of the electric field,

$$\begin{aligned} \frac{1}{\sin \theta} [\partial_\theta(\sin \theta E_\phi) - \partial_\phi E_\theta] \\ = \frac{i\omega\mu}{r} \nabla_a^2(r\pi_m), \end{aligned}$$

supplies the algebraic one, so that at $r = a$

$$\frac{1}{r} \partial_r(r\pi_m) = \frac{1}{r} \partial_r(r\pi_m^d), \quad \mu \pi_m = \mu_d \pi_m^d.$$

Scattering from Dielectric Spheres (cont.)

These four relations are the decoupled conditions (4.5.15)–(4.5.16). With this assignment, the coefficients below take the form in (4.5.23)–(4.5.24). The matching conditions ensure that each Debye potential outside the sphere couples exclusively to its counterpart inside, so each azimuthal dependence ($\cos \phi$ or $\sin \phi$) of the incident field is preserved in the scattered and internal fields. Thus, for the scattered field potential, we let

$$\pi_e^s = \frac{-E_0 \cos \phi}{k^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} a_n \hat{h}_n^{(2)}(kr) P_n^1(\cos \theta), \quad (4.5.17)$$

$$\pi_m^s = \frac{-E_0 \sin \phi}{\eta k^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} b_n \hat{h}_n^{(2)}(kr) P_n^1(\cos \theta), \quad (4.5.18)$$

Scattering from Dielectric Spheres (cont.)

where the Riccati–Hankel function $\hat{h}_n^{(2)}(kr) = kr h_n^{(2)}(kr)$ is used to meet the radiation condition, and the coefficients a_n and b_n are to be determined. The total potential outside the sphere is thus written as

$$\pi_e = \frac{E_0 \cos \phi}{k^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} \left[\hat{j}_n(kr) - a_n \hat{h}_n^{(2)}(kr) \right] P_n^1(\cos \theta), \quad (4.5.19)$$

and

$$\pi_m = \frac{E_0 \sin \phi}{\eta k^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} \left[\hat{j}_n(kr) - b_n \hat{h}_n^{(2)}(kr) \right] P_n^1(\cos \theta). \quad (4.5.20)$$

Scattering from Dielectric Spheres (cont.)

For the Debye potential inside the sphere, we let

$$\pi_e^d = \frac{E_0 \cos \phi}{k_d^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} c_n \hat{j}_n(k_d r) P_n^1(\cos \theta), \quad (4.5.21)$$

$$\pi_m^d = \frac{E_0 \sin \phi}{\eta_d k_d^2 r} \sum_{n=1}^{\infty} (-i)^{n-1} \frac{2n+1}{n(n+1)} d_n \hat{j}_n(k_d r) P_n^1(\cos \theta). \quad (4.5.22)$$

Applying the matching conditions (4.5.15) and (4.5.16) to the above equations mode by mode fixes the four unknown series. For the electric potential, the exterior and interior radial dependences per mode are (the common factor $E_0 \cos \phi (-i)^{n-1} \frac{2n+1}{n(n+1)} P_n^1(\cos \theta)$ cancels)

$$r\pi_e \sim \frac{1}{k^2} [\hat{j}_n(kr) - a_n \hat{h}_n^{(2)}(kr)], \quad r\pi_e^d \sim \frac{c_n}{k_d^2} \hat{j}_n(k_d r).$$

Scattering from Dielectric Spheres (cont.)

Continuity of $\frac{1}{r}\partial_r(r\pi_e)$ and of $\epsilon\pi_e$ from (4.5.15) then gives, at $r = a$,

$$\begin{aligned}\frac{1}{k} [\hat{j}'_n(ka) - a_n \hat{h}_n^{(2)'}(ka)] &= \frac{1}{k_d} c_n \hat{j}'_n(k_d a), \\ \frac{\epsilon}{k^2} [\hat{j}_n(ka) - a_n \hat{h}_n^{(2)}(ka)] &= \frac{\epsilon_d}{k_d^2} c_n \hat{j}_n(k_d a).\end{aligned}$$

Here and below, a prime on a Riccati–Bessel or Riccati–Hankel function means differentiation with respect to that function's complete dimensionless argument. To display the elimination explicitly, abbreviate

$$x = ka, \quad y = k_d a, \quad A = \sqrt{\epsilon\mu_d}, \quad B = \sqrt{\epsilon_d\mu},$$

and write $\hat{j}_x = \hat{j}_n(x)$, $\hat{h}_x = \hat{h}_n^{(2)}(x)$, and $\hat{j}_y = \hat{j}_n(y)$; here $x = ka$ is the Mie size parameter, distinct from the Legendre argument $x = \cos\theta$ used earlier. Since $\epsilon/k^2 = 1/(\omega^2\mu)$ and $\epsilon_d/k_d^2 = 1/(\omega^2\mu_d)$, the two electric equations above are equivalently

$$\hat{j}'_x - a_n \hat{h}'_x = c_n \frac{k}{k_d} \hat{j}'_y, \quad \hat{j}_x - a_n \hat{h}_x = c_n \frac{\mu}{\mu_d} \hat{j}_y.$$

Scattering from Dielectric Spheres (cont.)

To avoid dividing by either $\hat{j}_y$ or $\hat{j}'_y$, which can vanish at isolated arguments, cross-multiply the two equations. Using

$$\frac{\mu_d}{\mu} \frac{k}{k_d} = \sqrt{\frac{\epsilon \mu_d}{\epsilon_d \mu}} = \frac{A}{B}$$

shows that $B(k/k_d) = A(\mu/\mu_d)$. Multiplying the first equation by $B\hat{j}_y$ and the second by $A\hat{j}'_y$ therefore gives

$$B(\hat{j}'_x - a_n \hat{h}'_x) \hat{j}_y = A(\hat{j}_x - a_n \hat{h}_x) \hat{j}'_y.$$

Collecting the terms proportional to a_n therefore yields

$$a_n (A \hat{h}_x \hat{j}'_y - B \hat{h}'_x \hat{j}_y) = A \hat{j}_x \hat{j}'_y - B \hat{j}_x \hat{j}'_y,$$

Scattering from Dielectric Spheres (cont.)

which is (4.5.23). The internal coefficient can be obtained without cancelling an internal Bessel factor. Put $q = k/k_d$ and $s = \mu/\mu_d$, so $s/q = B/A$. The two electric equations are the 2×2 system

$$\begin{bmatrix} \hat{h}'_x & q\hat{j}'_y \\ \hat{h}_x & s\hat{j}_y \end{bmatrix} \begin{bmatrix} a_n \\ c_n \end{bmatrix} = \begin{bmatrix} \hat{j}'_x \\ \hat{j}_x \end{bmatrix}.$$

The Riccati–Bessel Wronskian is

$$\hat{j}_n(x)\hat{h}_n^{(2)'}(x) - \hat{j}_n'(x)\hat{h}_n^{(2)}(x) = -i.$$

If $D_e = A\hat{h}_x\hat{j}'_y - B\hat{h}'_x\hat{j}_y$ denotes the denominator of a_n , the system determinant is

$$\begin{aligned} \Delta_e &= s\hat{h}'_x\hat{j}_y - q\hat{h}_x\hat{j}'_y \\ &= -\frac{q}{A} (A\hat{h}_x\hat{j}'_y - B\hat{h}'_x\hat{j}_y) = -\frac{q}{A} D_e. \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

Cramer's rule and the Wronskian give the numerator $\hat{h}'_x \hat{j}_x - \hat{h}_x \hat{j}'_x = -i$, and hence

$$c_n = \frac{-i}{\Delta_e} = \frac{iA}{qD_e} = \frac{i\mu_d \sqrt{\epsilon_d/\mu}}{D_e},$$

which is (4.5.25).

For the magnetic potential, retaining its different normalization $1/(\eta k^2)$ is essential. Conditions (4.5.16) reduce mode by mode to

$$\hat{j}'_x - b_n \hat{h}'_x = d_n \frac{\mu}{\mu_d} \hat{j}'_y, \quad \hat{j}_x - b_n \hat{h}_x = d_n \frac{k}{k_d} \hat{j}_y.$$

Eliminating d_n and using $(k_d/k)(\mu/\mu_d) = B/A$ gives

$$A(\hat{j}'_x - b_n \hat{h}'_x) \hat{j}_y = B(\hat{j}_x - b_n \hat{h}_x) \hat{j}'_y,$$

so

$$b_n (B \hat{h}_x \hat{j}'_y - A \hat{h}'_x \hat{j}_y) = B \hat{j}_x \hat{j}'_y - A \hat{j}'_x \hat{j}_y,$$

Scattering from Dielectric Spheres (cont.)

which is (4.5.24). With $D_m = B\hat{h}_x\hat{j}'_y - A\hat{h}'_x\hat{j}_y$, retain the same $q = k/k_d$ and $s = \mu/\mu_d$. The magnetic equations form

$$\begin{bmatrix} \hat{h}'_x & s\hat{j}'_y \\ \hat{h}_x & q\hat{j}_y \end{bmatrix} \begin{bmatrix} b_n \\ d_n \end{bmatrix} = \begin{bmatrix} \hat{j}'_x \\ \hat{j}_x \end{bmatrix}.$$

Its determinant and Cramer's numerator are

$$\Delta_m = -\frac{q}{A}D_m, \quad \hat{h}'_x\hat{j}_x - \hat{h}_x\hat{j}'_x = -i,$$

so

$$d_n = \frac{-i}{\Delta_m} = \frac{iA}{qD_m} = \frac{i\mu_d\sqrt{\epsilon_d/\mu}}{D_m},$$

which is (4.5.26). These determinant formulas remain valid when $\hat{j}_y$ or $\hat{j}'_y$ vanishes individually. They assume $D_e \neq 0$ and $D_m \neq 0$; a zero of a full determinant is a genuine modal pole and must be handled by a limiting (usually lossy or complex-frequency) formulation. As an immediate consistency check, if $\epsilon_d = \epsilon$ and $\mu_d = \mu$, then $y = x$,

Scattering from Dielectric Spheres (cont.)

both scattering numerators vanish, and the Wronskian gives $a_n = b_n = 0$ and $c_n = d_n = 1$, as required when the “sphere” is indistinguishable from its host. The resulting coefficients are

$$a_n = \frac{\sqrt{\epsilon_d \mu_d} \hat{j}_n(ka) \hat{j}_n'(k_d a) - \sqrt{\epsilon_d \mu_d} \hat{j}_n'(ka) \hat{j}_n(k_d a)}{\sqrt{\epsilon_d \mu_d} \hat{h}_n^{(2)}(ka) \hat{j}_n'(k_d a) - \sqrt{\epsilon_d \mu_d} \hat{h}_n^{(2)'}(ka) \hat{j}_n(k_d a)}, \quad (4.5.23)$$

$$b_n = \frac{\sqrt{\epsilon_d \mu_d} \hat{j}_n(ka) \hat{j}_n'(k_d a) - \sqrt{\epsilon_d \mu_d} \hat{j}_n'(ka) \hat{j}_n(k_d a)}{\sqrt{\epsilon_d \mu_d} \hat{h}_n^{(2)}(ka) \hat{j}_n'(k_d a) - \sqrt{\epsilon_d \mu_d} \hat{h}_n^{(2)'}(ka) \hat{j}_n(k_d a)}, \quad (4.5.24)$$

Scattering from Dielectric Spheres (cont.)

$$c_n = \frac{i\mu_d \sqrt{\epsilon_d/\mu}}{\sqrt{\epsilon\mu_d} \hat{h}_n^{(2)}(ka) \hat{j}'_n(k_d a) - \sqrt{\epsilon_d\mu} \hat{h}_n^{(2)'}(ka) \hat{j}_n(k_d a)}, \quad (4.5.25)$$

$$d_n = \frac{i\mu_d \sqrt{\epsilon_d/\mu}}{\sqrt{\epsilon_d\mu} \hat{h}_n^{(2)}(ka) \hat{j}'_n(k_d a) - \sqrt{\epsilon\mu_d} \hat{h}_n^{(2)'}(ka) \hat{j}_n(k_d a)}. \quad (4.5.26)$$

Consider the far field at each fixed integer order n . The exact finite spherical formula in Section B.5.2 gives $\hat{h}_n^{(2)}(kr) = i^{n+1} e^{-ikr} [1 + O_n((kr)^{-1})]$. Equivalently, writing it in the cylindrical notation at $\nu = n + \frac{1}{2}$ gives

$$H_{n+1/2}^{(2)}(kr) \approx \sqrt{\frac{2}{\pi kr}} e^{-i\left(kr - \frac{(n+1/2)\pi}{2} - \frac{\pi}{4}\right)}.$$

Scattering from Dielectric Spheres (cont.)

Multiplying by $\sqrt{\pi kr/2}$ cancels the radical prefactor, leaving only the phase. The half-integer order contributes a further $-\pi/4$ —one eighth of a cycle—bringing the total constant phase shift to $-\pi/2$,

$$-\frac{(n+1/2)\pi}{2} - \frac{\pi}{4} = -\frac{n\pi}{2} - \frac{\pi}{4} - \frac{\pi}{4} = -\frac{n\pi}{2} - \frac{\pi}{2},$$

so that

$$\begin{aligned}\hat{h}_n^{(2)}(kr) &\approx e^{-i(kr - \frac{n\pi}{2} - \frac{\pi}{2})} \\ &= e^{i\frac{n\pi}{2}} e^{i\frac{\pi}{2}} e^{-ikr} = i^n i e^{-ikr}.\end{aligned}$$

Using $e^{i\pi/2} = i$ and $e^{in\pi/2} = i^n$, the leading factor collapses to i^{n+1} , giving

$$\hat{h}_n^{(2)}(kr) = \sqrt{\frac{\pi kr}{2}} H_{n+1/2}^{(2)}(kr) \approx i^{n+1} e^{-ikr}, \quad (4.5.27)$$

Scattering from Dielectric Spheres (cont.)

we get

$$\Pi_e^s = r\pi_e^s \approx e^{-ikr} \frac{E_0 \cos \phi}{k^2} \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} a_n P_n^1(\cos \theta), \quad (4.5.28)$$

$$\Pi_m^s = r\pi_m^s \approx e^{-ikr} \frac{E_0 \sin \phi}{\eta k^2} \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} b_n P_n^1(\cos \theta). \quad (4.5.29)$$

The fixed-order approximation may be passed through these sums for each fixed admissible sphere problem, but not merely because it holds for each individual mode. Section B.5.2 derives an absolute majorant for the angular coefficients and a separate summable majorant for the exact scaled radial functions and their required derivatives, uniform for $r \geq b > a$. Dominated convergence then justifies the far-field limit and its radial derivative. If M_n denotes the established nonnegative angular majorant and $M_{n+1} \leq qM_n$ beyond the chosen truncation threshold, with $q < 1$, the omitted tail is at most $M_{N+1}/(1 - q)$ by (B.5.5). This uses the first omitted majorant, not the first actual

Scattering from Dielectric Spheres (cont.)

modal term, which may vanish while later terms do not. The constants depend on size and material parameters; the statement does not assert uniformity near a modal pole or in a singular material limit. Noting that in the far field

$$\partial_r \Pi_e^s \approx -ik \Pi_e^s, \quad \partial_r \Pi_m^s \approx -ik \Pi_m^s, \quad (4.5.30)$$

we can thus express E_θ and E_ϕ in the far zone explicitly. Substituting (4.4.12), replacing $\partial_r \rightarrow -ik$ through the far-field rule above, and using the angular functions π_n and τ_n defined below, the θ -component becomes

$$\begin{aligned} E_\theta &\approx \frac{-ik}{r} \partial_\theta \Pi_e^s - \frac{i\omega\mu}{r \sin \theta} \partial_\phi \Pi_m^s \\ &= \frac{-i E_0 \cos \phi}{k} \frac{e^{-ikr}}{r} \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} a_n \tau_n \\ &\quad - \frac{i\omega\mu}{\eta} \frac{E_0 \cos \phi}{k^2} \frac{e^{-ikr}}{r} \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} b_n \pi_n, \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

where $\partial_\phi(\sin \phi) = \cos \phi$ has been used in the magnetic term. Since $\eta = \sqrt{\mu/\epsilon}$ and $k = \omega\sqrt{\mu\epsilon}$, we have $\omega\mu/\eta = k$, so one power of k cancels and both sums carry the common factor $E_0 \cos \phi$. Collecting them,

$$E_\theta \approx \frac{-i E_0 \cos \phi}{k} \frac{e^{-ikr}}{r} \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} (a_n \tau_n + b_n \pi_n),$$

which is exactly $E_0 (e^{-ikr}/r) f_\theta$ with $f_\theta = -i \cos \phi S_2/k$. An identical reduction of (4.4.12), now with $\partial_\phi(\cos \phi) = -\sin \phi$, gives

$$\begin{aligned} E_\phi &\approx \frac{-ik}{r \sin \theta} \partial_\phi \Pi_e^s + \frac{i\omega\mu}{r} \partial_\theta \Pi_m^s \\ &= \frac{i E_0 \sin \phi}{k} \frac{e^{-ikr}}{r} \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} (a_n \pi_n + b_n \tau_n), \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

so that $f_\phi = i \sin \phi S_1/k$. Reading off the angular sums motivates the definitions S_1, S_2 and the functions π_n, τ_n below. In accordance with (4.1.3), E_0 multiplies the scattering amplitude and is *not* part of $\vec{f}$:

$$E_\theta \approx E_0 f_\theta(\theta, \phi) \frac{e^{-ikr}}{r}, \quad (4.5.31)$$

$$E_\phi \approx E_0 f_\phi(\theta, \phi) \frac{e^{-ikr}}{r}, \quad (4.5.32)$$

$$f_\theta(\theta, \phi) = -i \cos \phi \frac{S_2(\theta)}{k}, \quad (4.5.33)$$

$$f_\phi(\theta, \phi) = i \sin \phi \frac{S_1(\theta)}{k}, \quad (4.5.34)$$

$$S_1(\theta) = \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} [a_n \pi_n(\cos \theta) + b_n \tau_n(\cos \theta)], \quad (4.5.35)$$

Scattering from Dielectric Spheres (cont.)

$$S_2(\theta) = \sum_{n=1}^{\infty} \frac{2n+1}{n(n+1)} [a_n \tau_n(\cos \theta) + b_n \pi_n(\cos \theta)], \quad (4.5.36)$$

with

$$\pi_n(\cos \theta) = \frac{P_n^1(\cos \theta)}{\sin \theta}, \quad \tau_n(\cos \theta) = \frac{d}{d\theta} P_n^1(\cos \theta). \quad (4.5.37)$$

By (4.1.5), the differential cross section is

$$\sigma_d(\theta, \phi) = |\vec{f}(\theta, \phi)|^2 = \left| \cos \phi \frac{S_2(\theta)}{k} \right|^2 + \left| \sin \phi \frac{S_1(\theta)}{k} \right|^2, \quad (4.5.38)$$

and by (4.1.8), the scattering cross section is

$$\sigma_{sca} = \int_{4\pi} \sigma_d(\theta, \phi) d\Omega = \frac{\pi}{k^2} \int_0^\pi (|S_2(\theta)|^2 + |S_1(\theta)|^2) \sin \theta d\theta. \quad (4.5.39)$$

Scattering from Dielectric Spheres (cont.)

Computing the following:

$$\begin{aligned} |S_1(\theta)|^2 &= \sum_{n=1}^{\infty} \sum_{n'=1}^{\infty} \frac{2n+1}{n(n+1)} \frac{2n'+1}{n'(n'+1)} \\ &\times [a_n \pi_n a_{n'}^* \pi_{n'} + b_n \tau_n b_{n'}^* \tau_{n'} \\ &\quad + a_n \pi_n b_{n'}^* \tau_{n'} + b_n \tau_n a_{n'}^* \pi_{n'}], \end{aligned} \quad (4.5.40)$$

$$\begin{aligned} |S_2(\theta)|^2 &= \sum_{n=1}^{\infty} \sum_{n'=1}^{\infty} \frac{2n+1}{n(n+1)} \frac{2n'+1}{n'(n'+1)} \\ &\times [a_n \tau_n a_{n'}^* \tau_{n'} + b_n \pi_n b_{n'}^* \pi_{n'} \\ &\quad + a_n \tau_n b_{n'}^* \pi_{n'} + b_n \pi_n a_{n'}^* \tau_{n'}], \end{aligned} \quad (4.5.41)$$

Scattering from Dielectric Spheres (cont.)

Adding the two expansions and grouping by amplitude pair, the angular factors combine into $\pi_n \pi_{n'} + \tau_n \tau_{n'}$ for the like-amplitude products ($a_n a_{n'}^*$ and $b_n b_{n'}^*$) and into $\pi_n \tau_{n'} + \tau_n \pi_{n'}$ for the mixed products ($a_n b_{n'}^*$ and $b_n a_{n'}^*$):

$$\begin{aligned} \int_0^\pi (|S_1|^2 + |S_2|^2) \sin \theta \, d\theta &= \sum_{n=1}^{\infty} \sum_{n'=1}^{\infty} \frac{2n+1}{n(n+1)} \frac{2n'+1}{n'(n'+1)} \\ &\times \left[(a_n a_{n'}^* + b_n b_{n'}^*) \int_0^\pi (\pi_n \pi_{n'} + \tau_n \tau_{n'}) \sin \theta \, d\theta \right. \\ &\quad \left. + (a_n b_{n'}^* + b_n a_{n'}^*) \int_0^\pi (\pi_n \tau_{n'} + \tau_n \pi_{n'}) \sin \theta \, d\theta \right]. \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

These angular integrals satisfy the orthogonality relations

$$\int_0^\pi (\pi_n \pi_{n'} + \tau_n \tau_{n'}) \sin \theta d\theta = \begin{cases} 0, & n \neq n', \\ \frac{2n^2(n+1)^2}{2n+1}, & n = n', \end{cases} \quad (4.5.42)$$

$$\int_0^\pi (\pi_n \tau_{n'} + \tau_n \pi_{n'}) \sin \theta d\theta = 0.$$

The second relation eliminates every mixed term, while the first collapses the like-amplitude terms to the diagonal $n' = n$. For that diagonal, the coefficient simplifies as

$$\left(\frac{2n+1}{n(n+1)} \right)^2 \frac{2n^2(n+1)^2}{2n+1} = 2(2n+1),$$

Scattering from Dielectric Spheres (cont.)

and with $a_n a_n^* = |a_n|^2$ and $b_n b_n^* = |b_n|^2$ we obtain

$$\begin{aligned}\sigma_{\text{sca}} &= \frac{\pi}{k^2} \int_0^\pi (|S_1|^2 + |S_2|^2) \sin \theta \, d\theta \\ &= \frac{2\pi}{k^2} \sum_{n=1}^{\infty} (2n+1) (|a_n|^2 + |b_n|^2) .\end{aligned}\tag{4.5.43}$$

For a Rayleigh sphere, *both* dimensionless size parameters must be small:

$$|x| = ka \ll 1, \quad |y| = |k_d|a \ll 1.$$

The equality $|x| = ka$ uses the lossless-host choice $k > 0$; the modulus on y is essential when the passive interior wavenumber k_d is complex. Define the dimensionless material wavenumber ratio at the start of the derivation by

$$\kappa \equiv \frac{y}{x} = \frac{k_d}{k}, \quad y = \kappa x.$$

Scattering from Dielectric Spheres (cont.)

Thus κ is not the spherical-harmonic azimuthal order m . For fixed material parameters, assume that all electric and magnetic quasistatic multipole denominators are nonzero:

$$n\epsilon_d + (n+1)\epsilon \neq 0, \quad n\mu_d + (n+1)\mu \neq 0, \quad n = 1, 2, \dots$$

Then the multipoles with $n > 1$ are of higher order, and the $n = 1$ electric and magnetic terms give the leading scattering. If a material contrast is instead allowed to vary with x , while the remaining normalized material parameters stay bounded and κ remains finite and nonzero, a generic sufficient condition for retaining only the quasistatic part of the denominator is that its normalized detuning remain asymptotically large compared with x^2 :

$$\left| \frac{\epsilon_d + 2\epsilon}{\epsilon} \right| \gg x^2, \quad \left| \frac{\mu_d + 2\mu}{\mu} \right| \gg x^2.$$

The condition is also necessary precisely when the $O(x^2)$ dynamic correction to the denominator has a nonzero coefficient: at detuning $O(x^2)$, that correction is of the same order as the quasistatic part and must be retained. If the dynamic coefficient vanishes

Scattering from Dielectric Spheres (cont.)

for a particular material combination, the quasistatic balance persists to smaller detuning, down to the order of the first nonvanishing correction. These detuning conditions control the dipole approximations only; they do not exclude higher-multipole resonances when the contrasts vary with x . For fixed material contrast, with A and B as defined above, substituting the small-argument expansions

$$\begin{aligned}\hat{j}_1(z) &= \frac{z^2}{3} + O(z^4), & \hat{j}_1'(z) &= \frac{2z}{3} + O(z^3), \\ \hat{h}_1^{(2)}(x) &= \frac{i}{x} + O(x), & \hat{h}_1^{(2)'}(x) &= -\frac{i}{x^2} + O(1)\end{aligned}$$

Scattering from Dielectric Spheres (cont.)

into the $n = 1$ numerator and denominator of (4.5.23) gives every leading term explicitly:

$$\begin{aligned} N_a &= A \hat{j}_1(x) \hat{j}_1'(y) - B \hat{j}_1'(x) \hat{j}_1(y) \\ &= A \left(\frac{x^2}{3} \right) \left(\frac{2\kappa x}{3} \right) - B \left(\frac{2x}{3} \right) \left(\frac{\kappa^2 x^2}{3} \right) + O(x^5) \\ &= \frac{2\kappa x^3}{9} (A - B\kappa) + O(x^5), \\ D_a &= A \hat{h}_1^{(2)}(x) \hat{j}_1'(y) - B \hat{h}_1^{(2)'}(x) \hat{j}_1(y) \\ &= A \left(\frac{i}{x} \right) \left(\frac{2\kappa x}{3} \right) - B \left(-\frac{i}{x^2} \right) \left(\frac{\kappa^2 x^2}{3} \right) + O(x^2) \\ &= \frac{i\kappa}{3} (2A + B\kappa) + O(x^2). \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

Because the quasistatic denominator is nonzero by hypothesis, division is valid. Cancelling the common nonzero factor κ , and using $1/i = -i$, gives

$$\begin{aligned}a_1 &= \frac{N_a}{D_a} = \frac{2\kappa x^3(A - B\kappa)/9}{i\kappa(2A + B\kappa)/3} + O(x^5) \\&= \frac{2x^3}{3i} \frac{A - B\kappa}{2A + B\kappa} + O(x^5) \\&= \frac{2i}{3} x^3 \frac{B\kappa - A}{2A + B\kappa} + O(x^5).\end{aligned}$$

Scattering from Dielectric Spheres (cont.)

The magnetic coefficient follows from (4.5.24) with the same steps:

$$\begin{aligned}N_b &= B\hat{j}_1(x)\hat{j}'_1(y) - A\hat{j}'_1(x)\hat{j}_1(y) \\&= \frac{2\kappa x^3}{9}(B - A\kappa) + O(x^5), \\D_b &= B\hat{h}_1^{(2)}(x)\hat{j}'_1(y) - A\hat{h}_1^{(2)'}(x)\hat{j}_1(y) \\&= \frac{i\kappa}{3}(2B + A\kappa) + O(x^2), \\b_1 &= \frac{N_b}{D_b} = \frac{2x^3}{3i} \frac{B - A\kappa}{2B + A\kappa} + O(x^5) \\&= \frac{2i}{3}x^3 \frac{A\kappa - B}{2B + A\kappa} + O(x^5).\end{aligned}$$

Scattering from Dielectric Spheres (cont.)

It remains to remove the square roots without silently changing branches. The correlated roots specified after (4.5.1) obey

$$\kappa^2 = \frac{\epsilon_d \mu_d}{\epsilon \mu}, \quad \frac{A}{B} = \frac{\mu_d}{\mu \kappa}.$$

Consequently,

$$\frac{B\kappa}{A} = \frac{\kappa^2 \mu}{\mu_d} = \frac{\epsilon_d}{\epsilon}, \quad \frac{A\kappa}{B} = \frac{\mu_d}{\mu}.$$

Substitution in the two ratios therefore gives

$$\begin{aligned} a_1 &= \frac{2i}{3}(ka)^3 \frac{\epsilon_d - \epsilon}{\epsilon_d + 2\epsilon} + O((ka)^5), \\ b_1 &= \frac{2i}{3}(ka)^3 \frac{\mu_d - \mu}{\mu_d + 2\mu} + O((ka)^5). \end{aligned} \tag{4.5.44}$$

When the corresponding contrast numerator is nonzero, these expressions show that the leading dimensionless Mie coefficient scales as $(ka)^3$. If $\epsilon_d = \epsilon$ or $\mu_d = \mu$, the associated

Scattering from Dielectric Spheres (cont.)

cubic term vanishes and, if that coefficient is needed, its next nonzero order must be retained. If both equalities hold, the sphere is identical to its host and all scattering coefficients vanish exactly, as established above. Keeping the $n = 1$ terms in $\sigma_{\text{sca}} = \frac{2\pi}{k^2} \sum_n (2n+1)(|a_n|^2 + |b_n|^2)$ and substituting (4.5.44) gives the complete leading result under the fixed-contrast nonresonance assumptions above

$$\begin{aligned}\sigma_{\text{sca}} &= \frac{2\pi}{k^2} 3 (|a_1|^2 + |b_1|^2) + O(k^6 a^8) \\ &= \frac{8\pi}{3} k^4 a^6 \left[\left| \frac{\epsilon_d - \epsilon}{\epsilon_d + 2\epsilon} \right|^2 + \left| \frac{\mu_d - \mu}{\mu_d + 2\mu} \right|^2 \right] + O(k^6 a^8).\end{aligned}$$

The material ratios in brackets are dimensionless, so the prefactor $k^4 a^6$ has the required cross-section dimension of area. Thus the often-quoted ω^4 (or λ^{-4}) law additionally assumes that the host and particle material parameters are approximately frequency independent over the band of interest. For ordinary nonmagnetic molecular matter, this leading electric-dipole behavior qualitatively explains why air molecules scatter blue light

Scattering from Dielectric Spheres (cont.)

more strongly than red light, why direct sunlight becomes redder along the long atmospheric path near sunset, and why the lunar sky remains black in the absence of an atmosphere; illumination, molecular response, observation angle, and optical path also enter, so this is a qualitative application rather than a deduction from the homogeneous-sphere solution alone.

The scattering from a PEC sphere can be obtained as a limiting case of the general dielectric Mie solution. Specifically, with μ_d held finite, real, and positive, let the permittivity $\epsilon_d \rightarrow -i\infty$. The matching conditions then enforce that the tangential electric field vanishes at the sphere surface, and the Mie coefficients simplify considerably. To see how this limit acts on (4.5.23)–(4.5.26), note that as $\epsilon_d \rightarrow -i\infty$ the interior wavenumber $k_d = \omega\sqrt{\epsilon_d\mu_d}$ also diverges. In each coefficient the two competing terms scale as $\sqrt{\epsilon\mu_d}$ and $\sqrt{\epsilon_d\mu}$. Since $|\sqrt{\epsilon_d\mu}| \rightarrow \infty$ while $\sqrt{\epsilon\mu_d}$ stays finite, only the $\sqrt{\epsilon_d\mu}$ terms survive in the numerator and denominator of a_n and b_n . For a_n

Scattering from Dielectric Spheres (cont.)

these are the primed factors $\hat{j}'_n(ka)$ and $\hat{h}_n^{(2)'}(ka)$, sharing the common factor $\sqrt{\epsilon_d \mu} \hat{j}_n(k_d a)$,

$$a_n \rightarrow \frac{-\sqrt{\epsilon_d \mu} \hat{j}'_n(ka) \hat{j}_n(k_d a)}{-\sqrt{\epsilon_d \mu} \hat{h}_n^{(2)'}(ka) \hat{j}_n(k_d a)} = \frac{\hat{j}'_n(ka)}{\hat{h}_n^{(2)'}(ka)}.$$

For b_n the surviving terms instead carry the unprimed factors $\hat{j}_n(ka)$, $\hat{h}_n^{(2)}(ka)$ together with the common factor $\sqrt{\epsilon_d \mu} \hat{j}'_n(k_d a)$,

$$b_n \rightarrow \frac{\sqrt{\epsilon_d \mu} \hat{j}_n(ka) \hat{j}'_n(k_d a)}{\sqrt{\epsilon_d \mu} \hat{h}_n^{(2)}(ka) \hat{j}'_n(k_d a)} = \frac{\hat{j}_n(ka)}{\hat{h}_n^{(2)}(ka)},$$

so the diverging factors cancel and the finite results below remain. For the internal coefficients the numerator grows at most like $\sqrt{\epsilon_d}$, while the denominator carries $\hat{j}_n(k_d a)$ or $\hat{j}'_n(k_d a)$ with $k_d a \rightarrow \infty$. The branch $\epsilon_d \rightarrow -i\infty$ fixes $k_d a \rightarrow |k_d a| e^{-i\pi/4}$, so that, with

Scattering from Dielectric Spheres (cont.)

the $e^{+i\omega t}$ convention adopted here, $\hat{j}_n(k_d a)$ grows without bound; the denominator therefore dominates and

$$c_n \sim \frac{i\mu_d \sqrt{\epsilon_d/\mu}}{-\sqrt{\epsilon_d\mu} \hat{h}_n^{(2)'}(ka) \hat{j}_n(k_d a)} \rightarrow 0,$$
$$d_n \sim \frac{i\mu_d \sqrt{\epsilon_d/\mu}}{\sqrt{\epsilon_d\mu} \hat{h}_n^{(2)}(ka) \hat{j}_n'(k_d a)} \rightarrow 0.$$

Writing the regular function as half incoming plus half outgoing, the dominant term along the ray $\arg y = -\pi/4$ is $\hat{j}_n(y) \sim \frac{1}{2}(-i)^{n+1}e^{iy}$, so the logarithmic derivative stays finite, $\hat{j}_n'(y)/\hat{j}_n(y) \rightarrow i$; this legitimizes comparing only the $\sqrt{\epsilon_d\mu}$ prefactors above. The vanishing of c_n and d_n is a statement about the coefficients alone; they carry no observation-radius dependence. For the reconstructed internal fields the corresponding limit holds at each fixed interior point, since $\hat{j}_n(k_d r)/\hat{j}_n(k_d a) \sim e^{ik_d(r-a)} \rightarrow 0$ for $r < a$,

Scattering from Dielectric Spheres (cont.)

but not uniformly up to $r = a$: the boundary products $c_n \hat{j}_n(k_d a)$ and $d_n \hat{j}'_n(k_d a)$ stay finite and encode the PEC surface current. Collecting these limits gives

$$a_n = \frac{\hat{j}'_n(ka)}{\hat{h}_n^{(2)'}(ka)}, \quad (4.5.45)$$

$$b_n = \frac{\hat{j}_n(ka)}{\hat{h}_n^{(2)}(ka)}, \quad (4.5.46)$$

$$c_n = 0, \quad (4.5.47)$$

$$d_n = 0. \quad (4.5.48)$$

All coefficients of the internal fields vanish, so the PEC solution is a special case embedded within the more general dielectric formulation. To obtain the monostatic curve without skipping its angular reduction, first use the phase-free convention $P_n^1(\cos \theta) = \sin \theta P'_n(\cos \theta)$, where here $P'_n(u) = dP_n(u)/du$. Using u as the

Scattering from Dielectric Spheres (cont.)

Legendre-polynomial argument keeps it distinct from the Mie size parameter $x = ka$. The Legendre equation at $u = 1$, together with $P_n(1) = 1$, gives

$$-2P'_n(1) + n(n+1)P_n(1) = 0, \quad P'_n(1) = \frac{n(n+1)}{2}.$$

Parity, $P_n(-u) = (-1)^n P_n(u)$, then gives $P'_n(-1) = (-1)^{n+1} n(n+1)/2$. Consequently, at backscatter $\theta = \pi$,

$$\begin{aligned}\pi_n(-1) &= P'_n(-1) = (-1)^{n+1} \frac{n(n+1)}{2}, \\ \tau_n(-1) &= -P'_n(-1) = (-1)^n \frac{n(n+1)}{2}.\end{aligned}$$

Substitution into the angular sums gives

$$\begin{aligned}S_1(\pi) &= \frac{1}{2} \sum_{n=1}^{\infty} (2n+1)(-1)^n (b_n - a_n), \\ S_2(\pi) &= -S_1(\pi).\end{aligned}$$

Scattering from Dielectric Spheres (cont.)

Although the individual spherical unit vectors depend on the arbitrary azimuth at the pole, the squared amplitude in (4.1.5) does not:

$$\sigma_d(\pi) = \frac{|S_1(\pi)|^2}{k^2}.$$

Using $\sigma_{\text{mono}} = 4\pi\sigma_d(\pi)$ and writing the electrical size as $x = ka = 2\pi a/\lambda$ therefore yields the dimensionless result used for the plot,

$$\frac{\sigma_{\text{mono}}}{\pi a^2} = \frac{1}{x^2} \left| \sum_{n=1}^{\infty} (2n+1)(-1)^n [a_n - b_n] \right|^2, \quad x = ka. \quad (4.5.49)$$

Figure 4.1 plots (4.5.49) against $x = ka$, not against a/λ ; the conversion is $a/\lambda = x/(2\pi)$. The reference value πa^2 can be derived within physical optics rather than left as an unevaluated stationary-phase claim. This is an approximation model, not a rigorous error estimate for the exact Mie series.

Scattering from Dielectric Spheres (cont.)

Take incidence in $+z$, with $\vec{E}_i = E_0 \hat{x} e^{-ikz}$ and $\vec{H}_i = (E_0/\eta) \hat{y} e^{-ikz}$. Locally replacing the illuminated PEC surface by its tangent plane gives $\vec{H}_{\text{tot},t} \approx 2\vec{H}_{i,t}$; physical optics therefore uses $\vec{J}_{\text{PO}} = 2\hat{n} \times \vec{H}_i$ on the illuminated hemisphere $z < 0$ and zero current on the shadowed hemisphere. This local reflection rule is the physical-optics assumption. In the back direction $\hat{s} = -\hat{z}$, the transverse current is $-(2E_0/\eta)n_z e^{-ikz} \hat{x}$. Substitution into the electric-current far-field radiation integral gives

$$\vec{f}_{\text{PO}}(-\hat{z}, \hat{z}) = \frac{ik}{2\pi} \hat{x} \int_{z' < 0} n_z e^{-2ikz'} dS'.$$

On the illuminated sphere, $z' = -\sqrt{a^2 - \rho^2}$ and $n_z dS' = -\rho d\rho d\phi$. Hence, with $u = \sqrt{a^2 - \rho^2}$,

$$\begin{aligned} f_{\text{PO},x} &= -ik \int_0^a \rho e^{2ik\sqrt{a^2 - \rho^2}} d\rho \\ &= -ik \int_0^a u e^{2iku} du \\ &= -\frac{a}{2} e^{2ika} - \frac{i}{4k} (e^{2ika} - 1). \end{aligned}$$

Scattering from Dielectric Spheres (cont.)

The last term is bounded by $1/(2k)$, so

$$\sigma_{\text{mono,PO}} = 4\pi |f_{\text{PO},x}|^2 = \pi a^2 [1 + O((ka)^{-1})], \quad ka \rightarrow \infty.$$

This establishes why the physical-optics reference level in the chosen normalization tends to one. It neither alters the exact formula (4.5.49) nor bounds the difference between physical optics and the exact Mie solution, which also contains diffraction and creeping-wave contributions.

Scattering from Dielectric Spheres

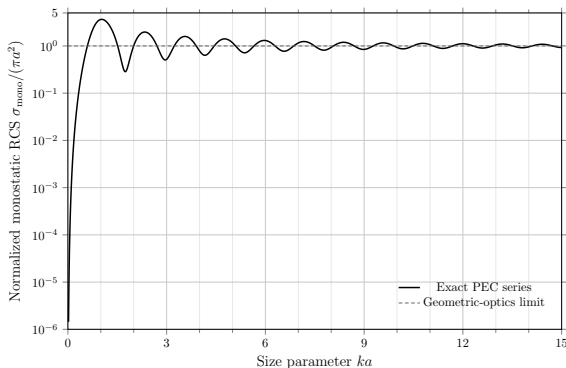


Figure 4.1: Normalized monostatic RCS $\sigma_{\text{mono}}/(\pi a^2)$ of a PEC sphere versus the size parameter $x = ka = 2\pi a/\lambda$.

Stratton curl modes and the Debye dictionary

Scattering computations are often written with Stratton's curl modes rather than with the Debye potentials above; the companion high-frequency volume, for example, expands its creeping-wave fields in that language. The two formulations generate exactly the same transverse-electric-to- $\hat{r}$ and transverse-magnetic-to- $\hat{r}$ mode families. This subsection records the translation once, including one sign convention that differs between the two presentations, so that any coefficient can be carried across without re-derivation.

Let ψ satisfy the scalar Helmholtz equation. With the phase-free P_n^m convention used throughout this chapter, the regular scalar seeds of azimuthal order m are

$$\begin{aligned}\psi_{emn} &= j_n(kr) P_n^m(\cos \theta) \cos m\phi, \\ \psi_{omn} &= j_n(kr) P_n^m(\cos \theta) \sin m\phi,\end{aligned}\tag{4.5.50}$$

with $n \geq 1$; replacing j_n by the outgoing $h_n^{(2)}$ of (4.4.29) gives the radiating seeds. The Stratton modes are the curls

$$\vec{M}_\zeta = \vec{\nabla} \times (\vec{r} \psi_\zeta), \quad \vec{N}_\zeta = \frac{1}{k} \vec{\nabla} \times \vec{M}_\zeta,\tag{4.5.51}$$

Stratton curl modes and the Debye dictionary (cont.)

where ζ stands for any $(e/o)mn$ label and $\vec{r}$ is the position vector. Divergences of curls vanish identically, so both families are solenoidal with no further computation. Moreover $\vec{\nabla} \times \vec{\nabla} \times \vec{M}_\zeta = -\vec{\nabla}^2 \vec{M}_\zeta$, while $\vec{\nabla}^2$ commutes with $\vec{\nabla} \times$; since $\vec{\nabla}^2(\psi\vec{r}) = (\vec{\nabla}^2\psi)\vec{r} + 2\vec{\nabla}\psi = -k^2\psi\vec{r} + 2\vec{\nabla}\psi$, taking one more curl kills the gradient term:

$$-\vec{\nabla}^2 \vec{M}_\zeta = -\vec{\nabla} \times [-k^2\psi_\zeta\vec{r} + 2\vec{\nabla}\psi_\zeta] = k^2\vec{M}_\zeta.$$

Hence

$$\begin{aligned}\vec{\nabla} \times \vec{\nabla} \times \vec{M}_\zeta &= k^2\vec{M}_\zeta, \\ \vec{\nabla} \times \vec{M}_\zeta &= k\vec{N}_\zeta, \\ \vec{\nabla} \times \vec{N}_\zeta &= k\vec{M}_\zeta.\end{aligned}\tag{4.5.52}$$

With $\eta = \omega\mu/k = k/(\omega\epsilon)$ from (4.5.1), the pairs $(\vec{E}, \vec{H}) = (\vec{M}_\zeta, (i/\eta)\vec{N}_\zeta)$ and $(\vec{E}, \vec{H}) = (\vec{N}_\zeta, (i/\eta)\vec{M}_\zeta)$ satisfy the source-free Maxwell curls mode by mode. Indeed,

$$\vec{\nabla} \times \vec{M}_\zeta = k\vec{N}_\zeta, \quad \vec{\nabla} \times [(i/\eta)\vec{N}_\zeta] = (i/\eta)k\vec{M}_\zeta,$$

because $\omega\mu/\eta = k$ and $k/\eta = \omega\epsilon$.

Stratton curl modes and the Debye dictionary (cont.)

Writing $\vec{\nabla} \times (\psi \vec{r}) = \vec{\nabla} \psi \times \vec{r}$ (since $\vec{\nabla} \times \vec{r} = \vec{0}$), the radial component is

$$\hat{r} \cdot \vec{M}_\zeta = \hat{r} \cdot (\vec{\nabla} \psi_\zeta \times \vec{r}) = 0$$

by the scalar triple product. Every $\vec{M}$ mode is therefore transverse to $\hat{r}$ (TE-to- $\hat{r}$); the companion $\vec{N}_\zeta$ carries the radial electric field (TM-to- $\hat{r}$), as (4.4.12a) shows. With $\Pi_m = C r \psi_\zeta$, so that $\Pi_m \hat{r} = C \psi_\zeta \vec{r}$, the magnetic-Debye field is

$$\vec{E}_m = -i\omega\mu\vec{\nabla} \times (\Pi_m \hat{r}) = -i\omega\mu C \vec{M}_\zeta,$$

TE-to- $\hat{r}$. With $\Pi_e = D r \psi_\zeta$, the electric-Debye field is

$$\vec{E}_e = \vec{\nabla} \times \vec{\nabla} \times (\Pi_e \hat{r}) = Dk \vec{N}_\zeta,$$

TM-to- $\hat{r}$. In compact form,

$$\begin{aligned} \vec{M}_\zeta &\text{ is TE-to-}\hat{r} \text{ (magnetic-Debye family),} \\ \vec{N}_\zeta &\text{ is TM-to-}\hat{r} \text{ (electric-Debye family).} \end{aligned} \tag{4.5.53}$$

Stratton curl modes and the Debye dictionary (cont.)

As a check, rebuild the incident x-polarized plane wave in this language. The projected potentials share the phasing $(-i)^{n-1}(2n+1)/[n(n+1)]$ with radial factors $\hat{j}_n/(k^2 r)$, with only $m=1$ surviving. Since $\hat{j}_n = kr j_n$, the electric potential gives

$$\Pi_e^i \hat{r} = (E_0/k) \sum_n (-i)^{n-1} (2n+1)/[n(n+1)] (\vec{r} \psi_{e1n})$$

with ψ_{e1n} from (4.5.50), whence

$$\vec{E}_e^i = E_0 \sum_n (-i)^{n-1} (2n+1)/[n(n+1)] \vec{N}_{e1n}^{(1)},$$

where (1) denotes regular j_n radial dependence. Likewise

$$\Pi_m^i \hat{r} = (E_0/(\eta k)) \sum_n (-i)^{n-1} (2n+1)/[n(n+1)] (\vec{r} \psi_{o1n}),$$

so that, because $\omega\mu/(\eta k) = 1$,

$$\begin{aligned} \vec{E}_m^i &= -i\omega\mu \vec{\nabla} \times (\Pi_m^i \hat{r}) \\ &= -iE_0 \sum_n (-i)^{n-1} (2n+1)/[n(n+1)] \vec{M}_{o1n}^{(1)}. \end{aligned}$$

Stratton curl modes and the Debye dictionary (cont.)

Since $-i \cdot (-i)^{n-1} = (-i)^n$, this is $E_0 \sum_n (-i)^n (2n+1)/[n(n+1)] \vec{M}_{o1n}^{(1)}$. Writing $(-i)^{n-1} = (-i)^n \cdot i$ casts the electric part in the same phasing, and both parts combine into

$$\vec{E}^i = E_0 \sum_{n=1}^{\infty} (-i)^n \frac{2n+1}{n(n+1)} [\vec{M}_{o1n}^{(1)} + i\vec{N}_{e1n}^{(1)}]. \quad (4.5.54)$$

This is the incident expansion used in the companion volume.

For the scattered field, watch one sign convention. The physical decomposition is total = incident + scattered. In this chapter the scattered Debye potential is parameterized with a minus sign multiplying a_n or b_n , so the total radial factor is regular minus coefficient times outgoing, (4.5.19)–(4.5.20): $\pi_e \sim [\hat{j}_n - a_n \hat{h}_n^{(2)}]$. The Stratton convention writes total = regular + (coefficient)(outgoing). Per mode this gives, with c_n^h the hard (TM-like) and c_n^s the soft (TE-like) Stratton coefficients,

$$c_n^h = -a_n, \quad c_n^s = -b_n, \quad (4.5.55)$$

with PEC forms $c_n^h = -\hat{j}_n/\hat{h}_n^{(2)'} and $c_n^s = -\hat{j}_n/\hat{h}_n^{(2)}$ from (4.5.45)–(4.5.46). Consequently $(c_n^h - c_n^s) = (b_n - a_n)$ in the PEC limit, the summand of (4.5.49) up to the overall sign, which the two far-field reductions carry consistently.$

Stratton curl modes and the Debye dictionary (cont.)

In short: a_n is TM (electric Debye) and b_n is TE (magnetic Debye), while the scattered sign differs, (4.5.55), because this chapter subtracts the outgoing term while the Stratton form adds it.

Further Reading

For a concise and accessible overview of special functions, the reader is encouraged to consult Novak and Fox [20]. A more comprehensive mathematical treatment can be found in Arfken, Weber, and Harris' *Mathematical Methods for Physicists* [1], which provides detailed derivations and properties of Bessel, Legendre, and other special functions widely used in physics and engineering. The *NIST Digital Library of Mathematical Functions (DLMF)* [9] serves as an authoritative online reference for the definitions, identities, and asymptotic expansions of special functions. Harrington's text [11] includes dedicated chapters on cylindrical and spherical wave solutions. In this work, we adopt the same selection of canonical examples—conducting cylinders and dielectric spheres—as in Zangwill's treatment [31], since these configurations serve as representative models for physical objects such as wires and raindrops. Ishimaru's classic [12] provides a rigorous formulation of Mie theory, which forms the principal reference for this chapter. For a broader discussion of electromagnetic scattering phenomena, the reader is referred to Osipov and Tretyakov's *Modern Electromagnetic Scattering Theory with Applications* [22]. In the context of light scattering by small particles, classical treatments include Van de Hulst's *Light Scattering by Small Particles* [29], Bohren and Huffman's *Absorption and Scattering of Light by Small Particles* [4], and Kerker's *Scattering of Light and Other Electromagnetic Radiation* [16].

- 1 Using (4.3.1) and (4.3.6), plot the well-defined normalized magnitude $|E_z^t(\rho, \phi)|/E_0$ versus $k\rho \geq 1$ for $ka = 1$ along the fixed rays $\phi = 0, \pi/2$, and π . State the series truncation used and verify numerically that $E_z^t = 0$ at $k\rho = ka = 1$. Distinguish series-truncation error from floating-point rounding: raising the truncation past convergence need not reduce the error once the rounding floor is reached.
- 2 Verify directly that (4.3.12)–(4.3.14) satisfy Faraday's law and the source-free Ampère–Maxwell law term by term.
- 3 For $ka = 1$, evaluate the TM surface current from both (4.3.15) and (4.3.16); demonstrate numerical agreement for a stated series truncation.
- 4 For the TE solution (4.3.24), evaluate (4.3.27) at $ka = 1$ and verify numerically that $E_\phi^t(a, \phi) = 0$.
- 5 Independently reproduce the mode-by-mode elimination leading to (4.5.23)–(4.5.26), including the Riccati–Bessel Wronskian used for c_n and d_n .
- 6 Verify (4.5.44).

Problems (cont.)

- 7 Verify (4.5.54) numerically: with $k = 1$ and $E_0 = 1$, evaluate the curl definitions (4.5.51) by finite differences (or explicit components) with the phase-free P_n^1 , sum $n = 1$ through 12 at $(x, y, z) = (0.5, 0.2, 0.3)$, and confirm agreement with $\hat{x}e^{-ikz}$ to 10^{-3} . State how the error changes when the truncation is raised to $n = 16$. A finite-difference implementation additionally needs a stated step size with a differentiation-convergence study, since truncation, differencing, and rounding errors behave differently.

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